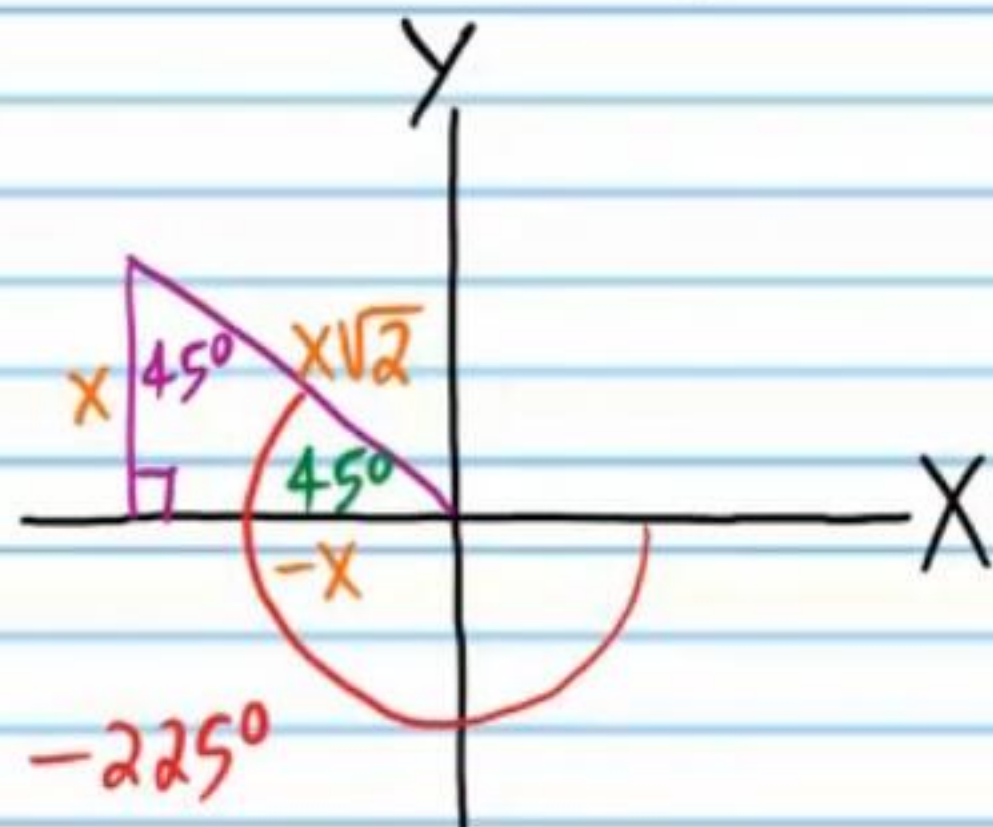
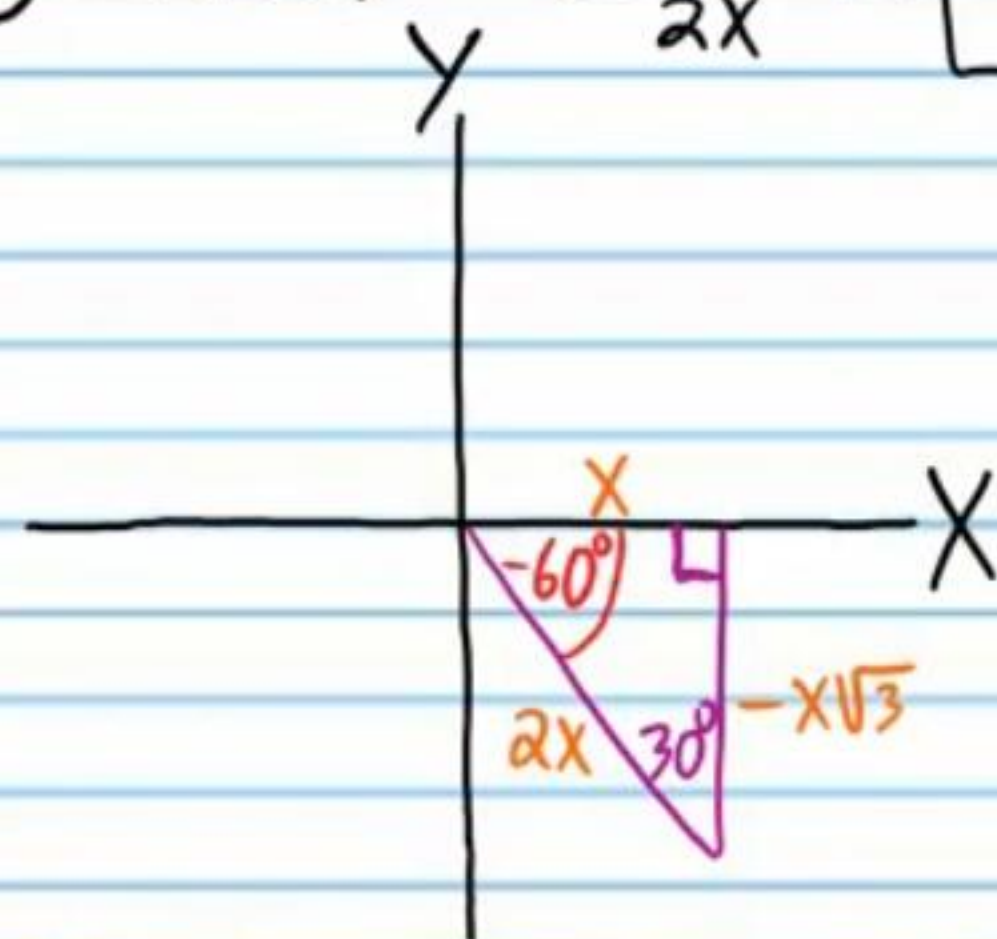


Continued Exploration of the Six Basic Trigonometric Functions

Negative Angle Measures

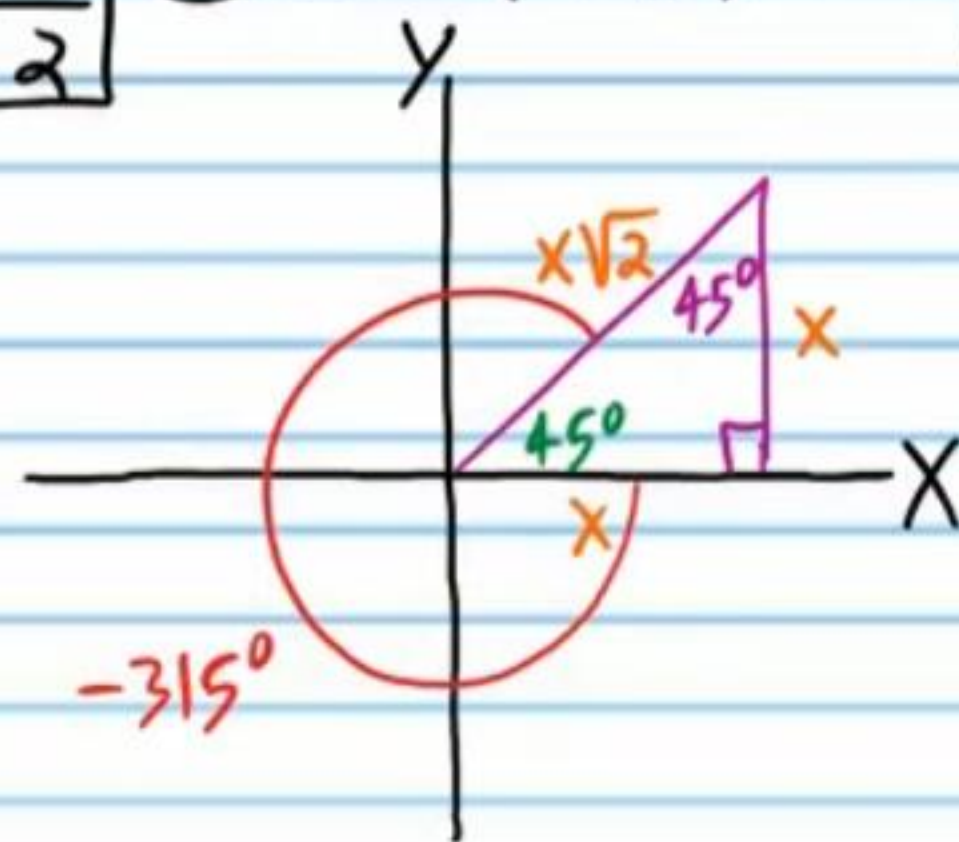
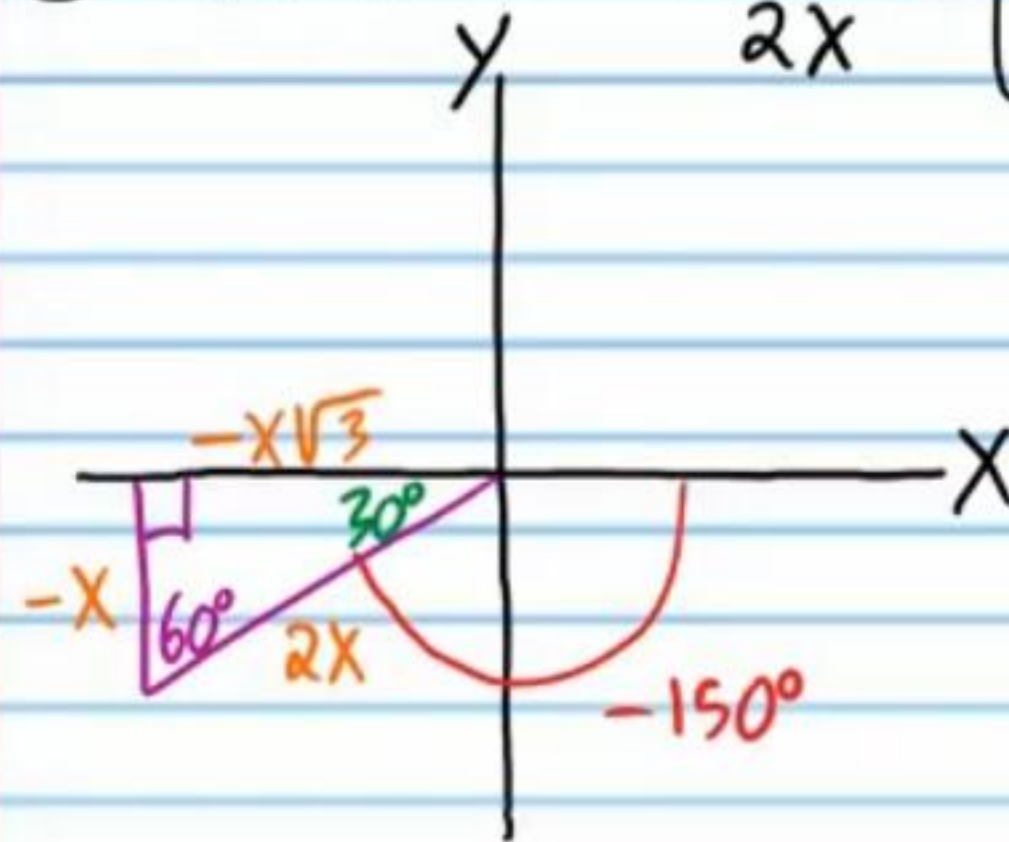
Draw each angle in standard position, write the measure of the reference angle, and draw a triangle to help find the value of the trigonometric expression. Then, find the value of the trigonometric expression *without a calculator*.

• ① $\sin(-60^\circ) = \frac{-x\sqrt{3}}{2x} = \boxed{-\frac{\sqrt{3}}{2}}$ ② $\cot(-225^\circ) = \frac{-x}{x} = \boxed{-1}$

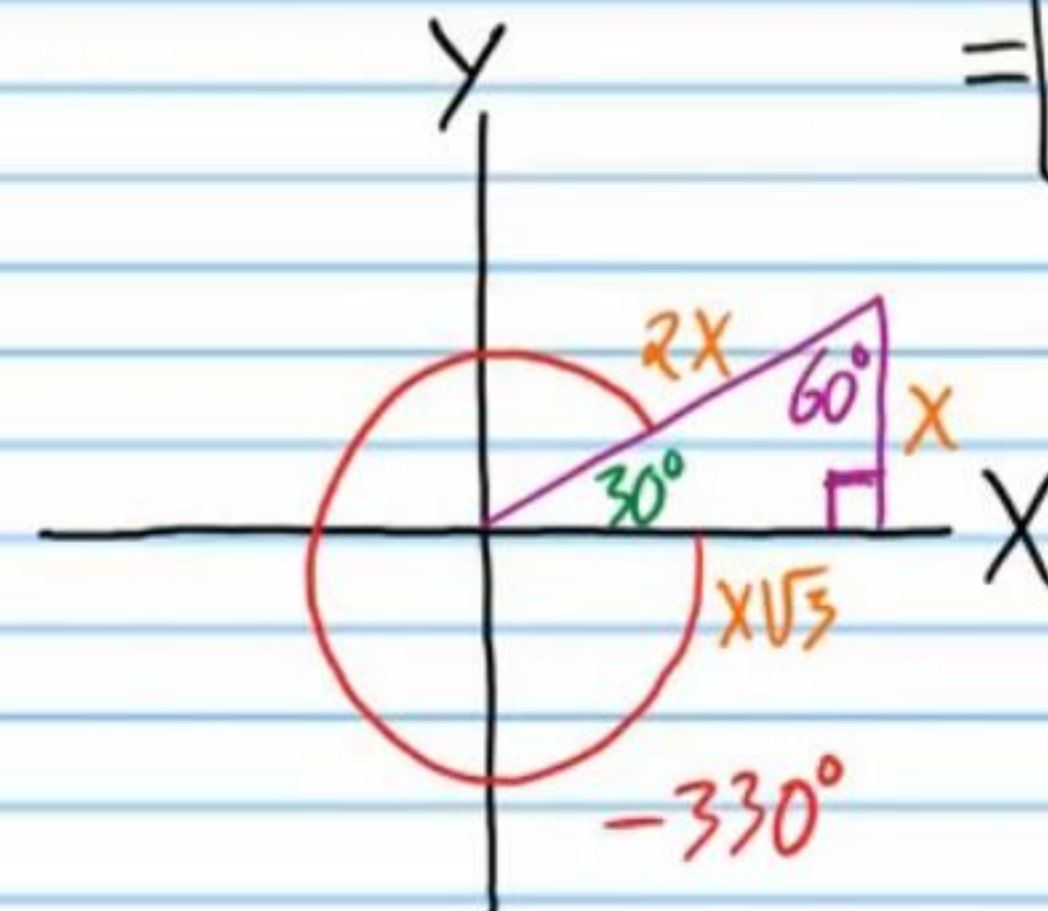
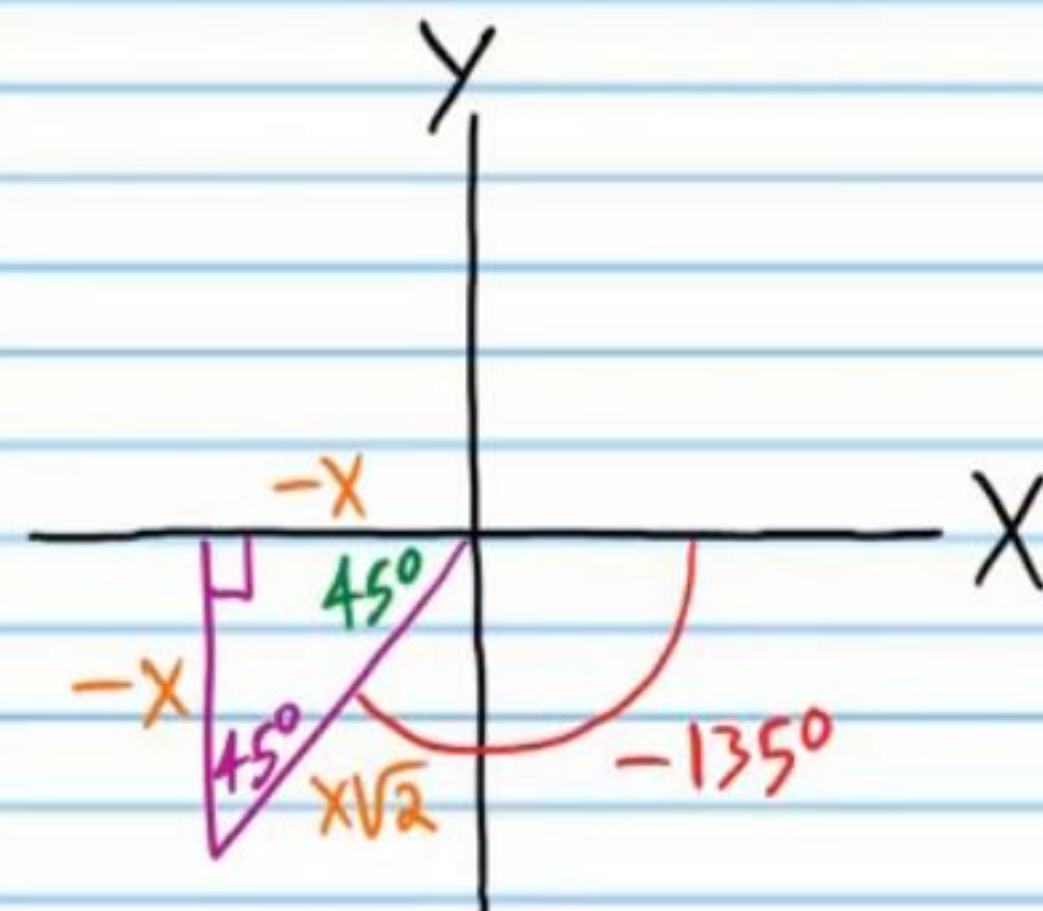


The Reference Angle of An Angle in Standard Position: The acute angle between the x-axis and the terminal side.

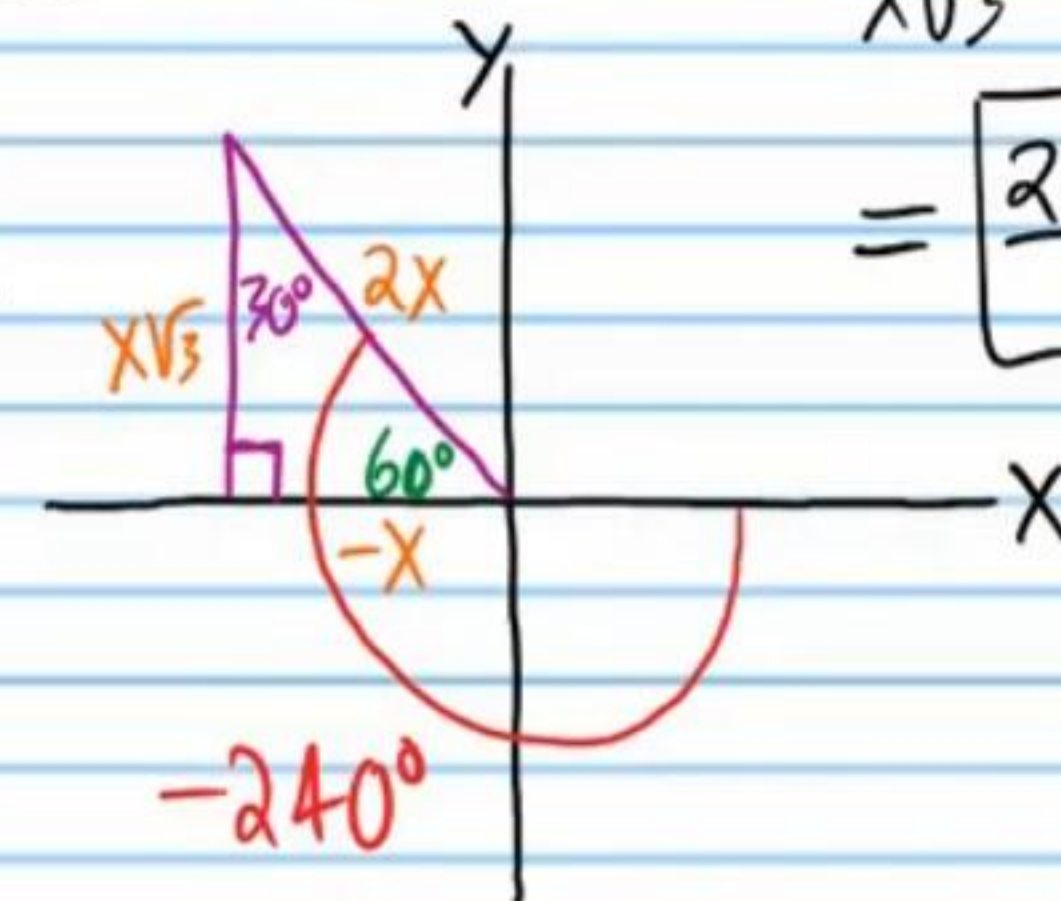
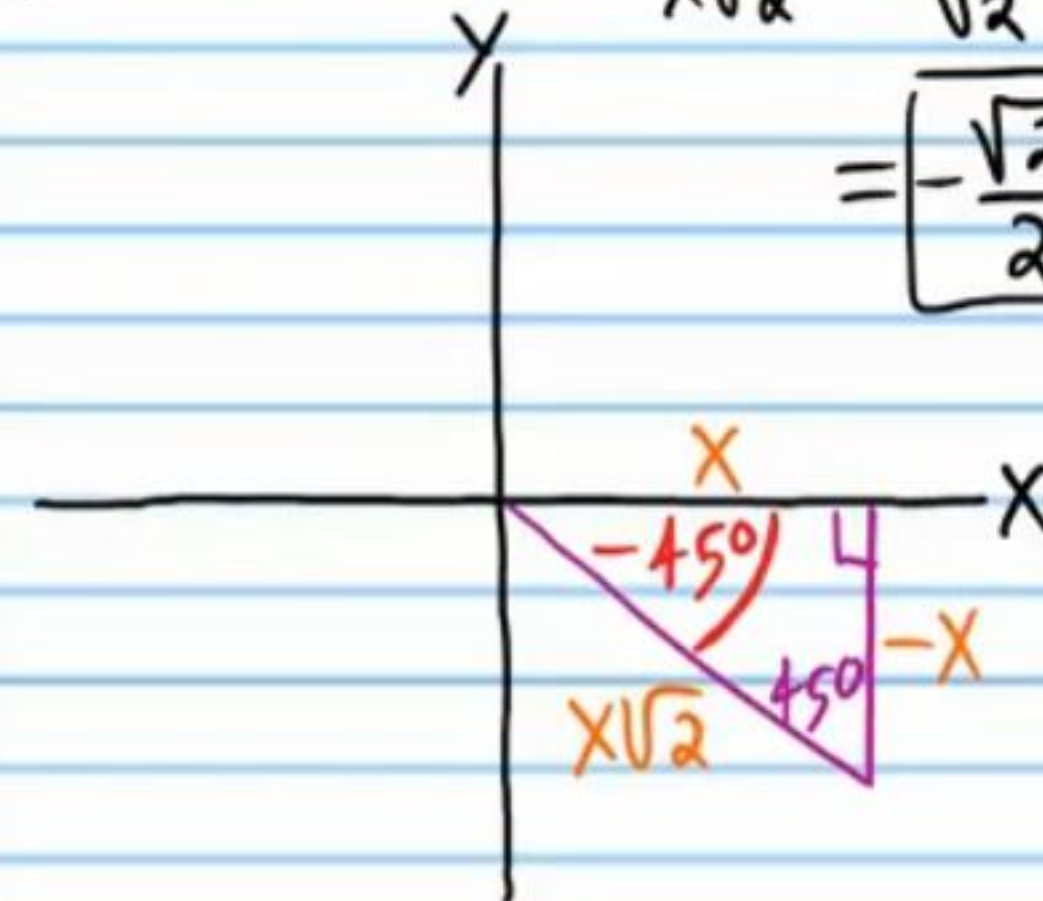
③ $\cos(-150^\circ) = \frac{-x\sqrt{3}}{2x} = \boxed{-\frac{\sqrt{3}}{2}}$ ④ $\csc(-315^\circ) = \frac{x\sqrt{2}}{x} = \boxed{\sqrt{2}}$



• ⑤ $\tan(-135^\circ) = \frac{-x}{-x} = \boxed{1}$ ⑥ $\sec(-330^\circ) = \frac{2x}{x\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$

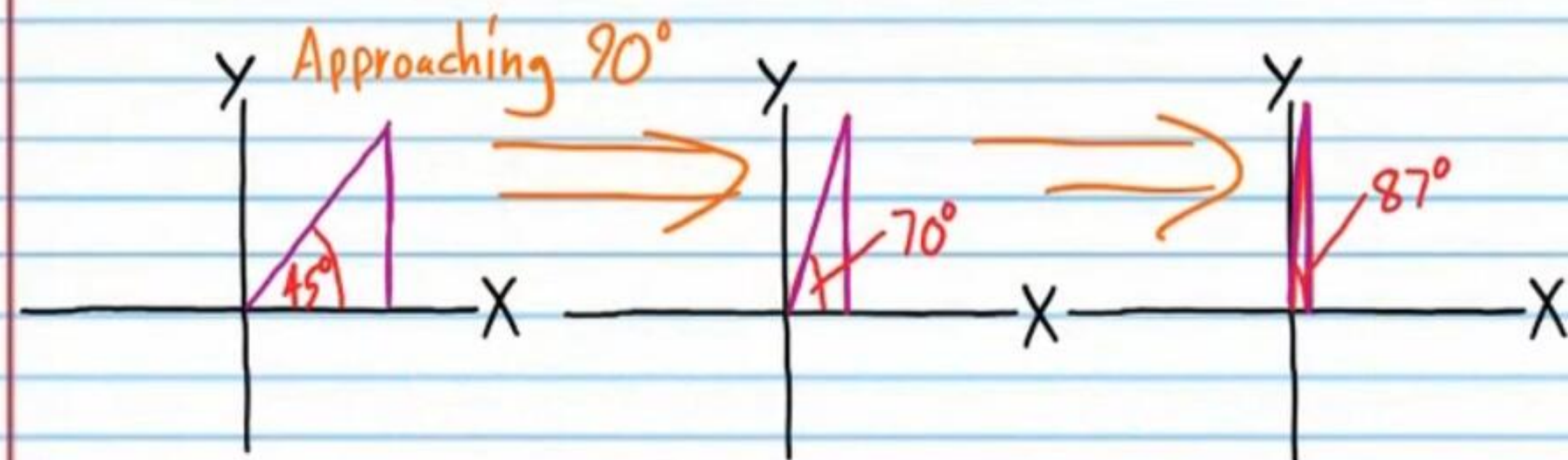


⑦ $\sin(-45^\circ) = \frac{-x}{x\sqrt{2}} = \frac{-1}{\sqrt{2}} = \boxed{-\frac{\sqrt{2}}{2}}$ ⑧ $\csc(-240^\circ) = \frac{2x}{x\sqrt{3}} = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}}$

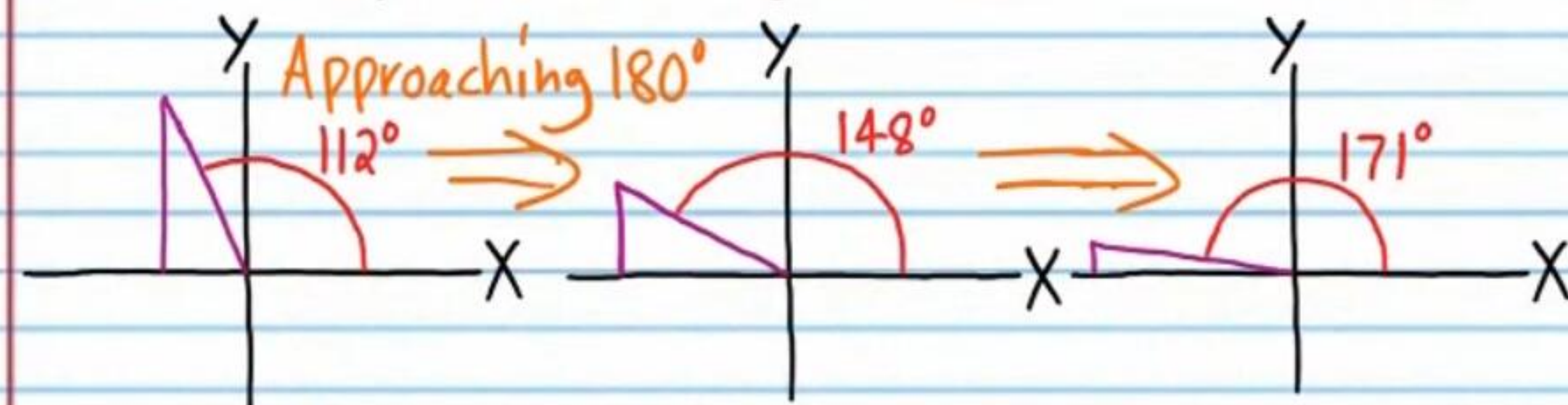


Collapsed Triangles ($\theta = 90k$, k any integer)

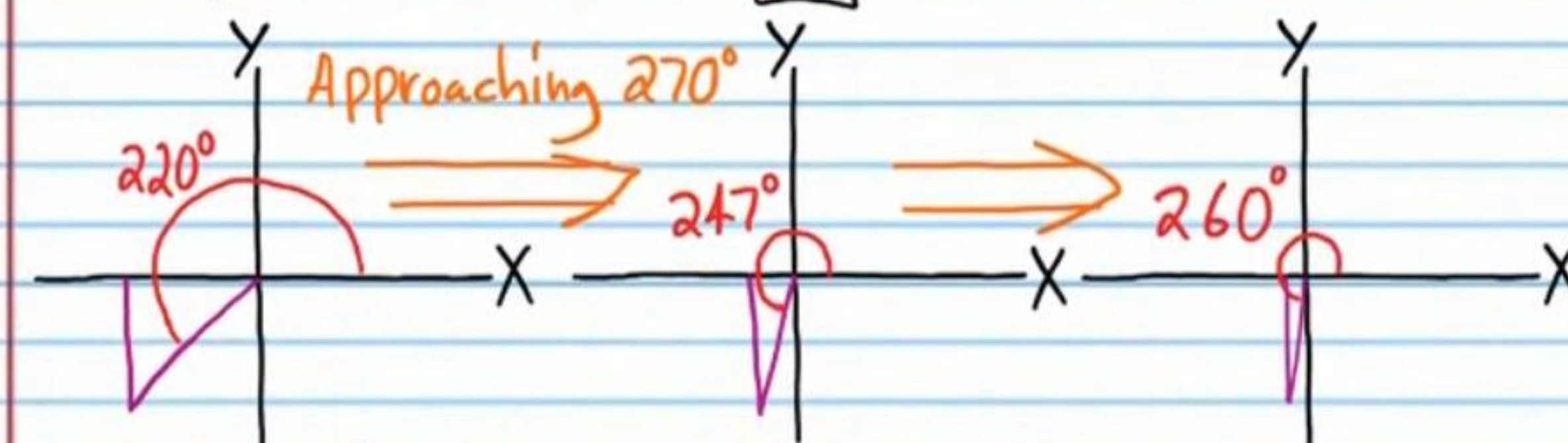
Example I) $\sin 90^\circ = \boxed{1}$



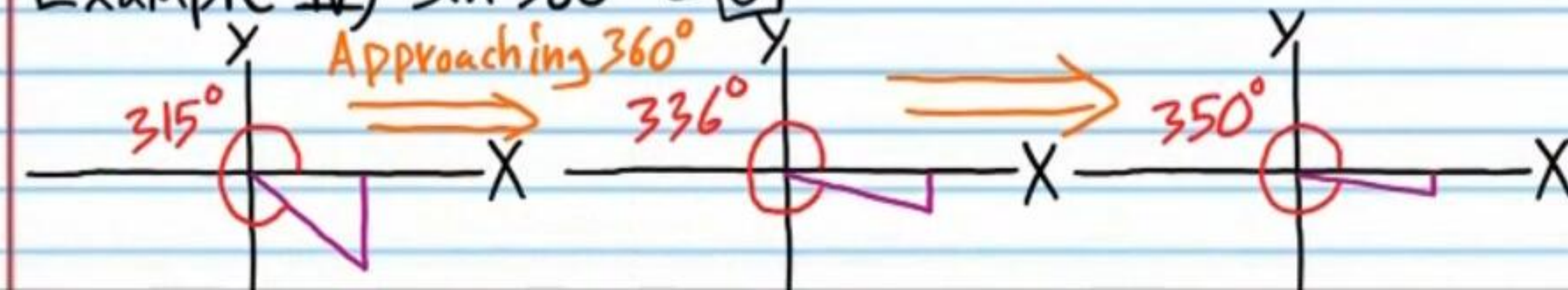
Example II) $\sin 180^\circ = \boxed{0}$



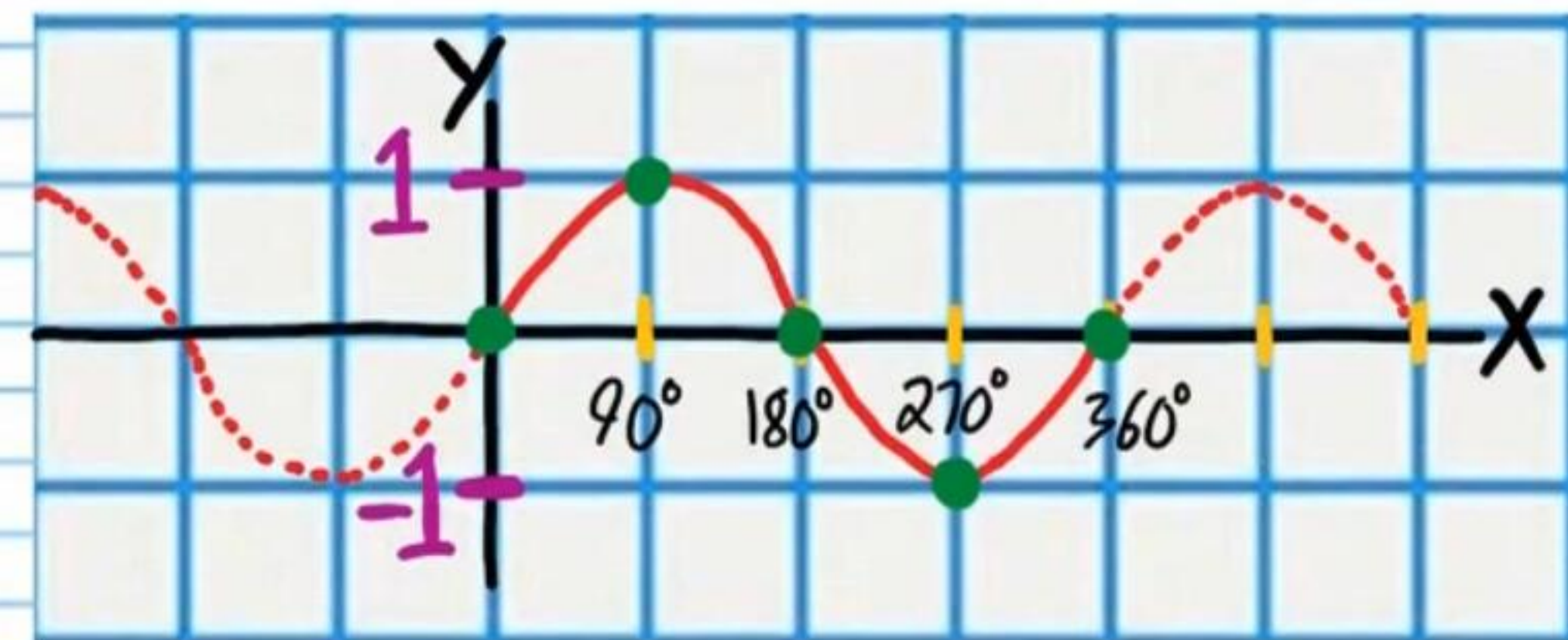
Example III) $\sin 270^\circ = \boxed{-1}$



Example IV) $\sin 360^\circ = \boxed{0}$



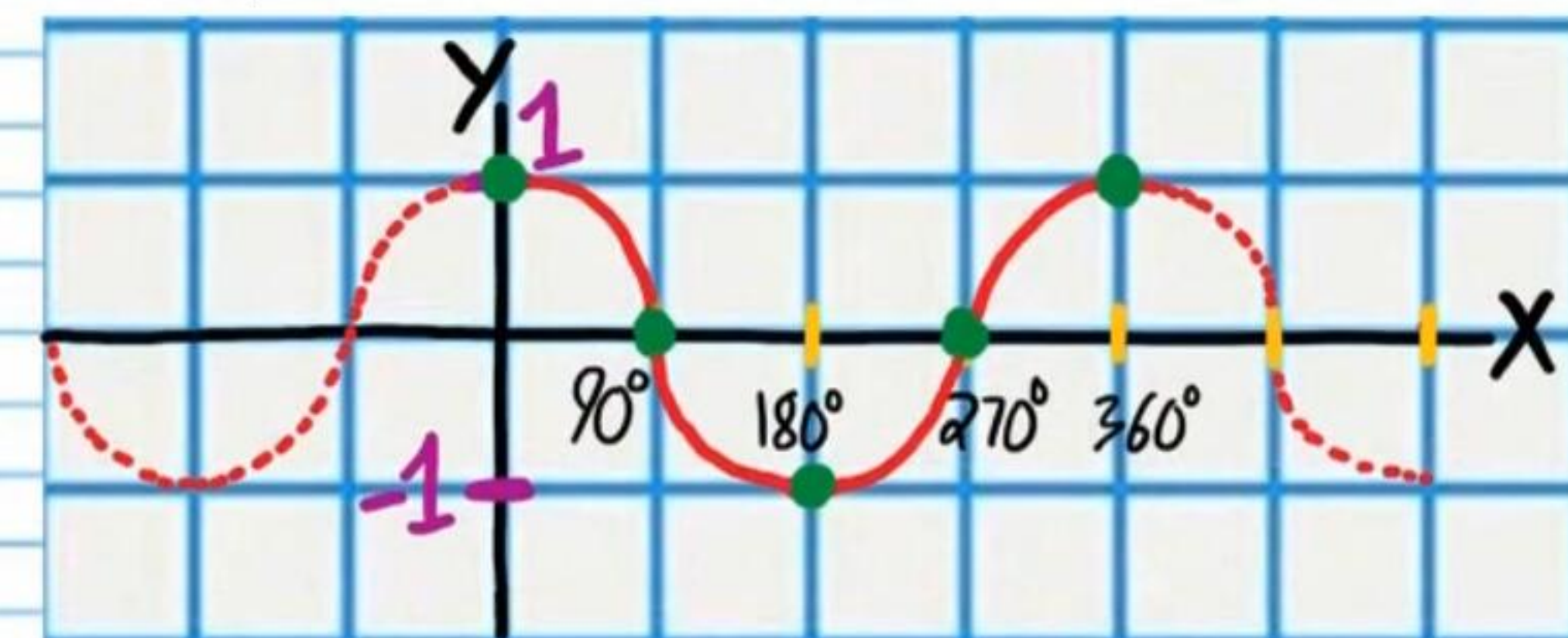
$$y = \sin x$$



Use the graph above to find the following values.

- ⑨ $\sin 270^\circ = \boxed{-1}$ ⑩ $\sin 360^\circ = \boxed{0}$ ⑪ $\sin 0^\circ = \boxed{0}$
 ⑫ $\sin(-90^\circ) = \boxed{-1}$ ⑬ $\sin(-180^\circ) = \boxed{0}$ ⑭ $\sin 450^\circ = \boxed{1}$

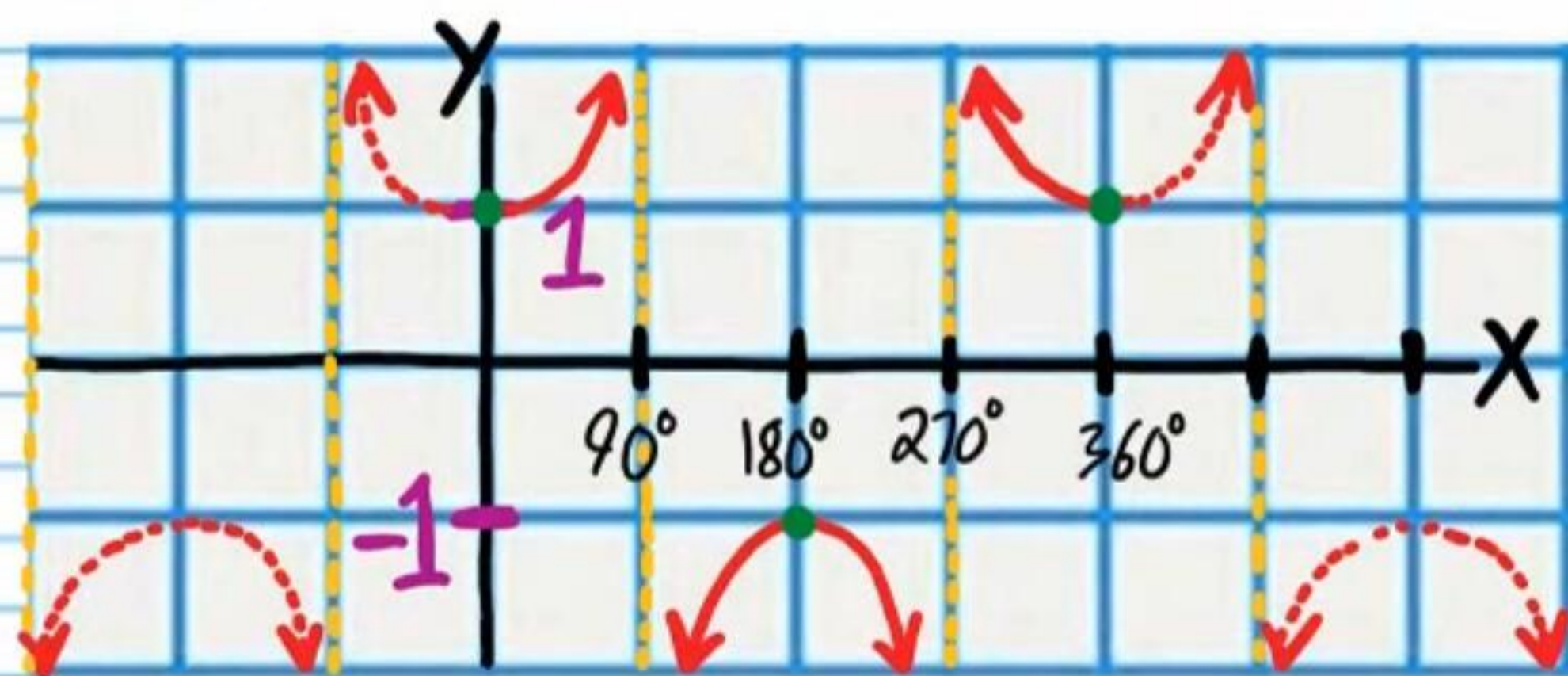
$$y = \cos x$$



Use the graph above to find the following values.

- ⑮ $\cos 90^\circ = \boxed{0}$ ⑯ $\cos 180^\circ = \boxed{-1}$ ⑰ $\cos 270^\circ = \boxed{0}$
 ⑱ $\cos 360^\circ = \boxed{1}$ ⑲ $\cos 0^\circ = \boxed{1}$ ⑳ $\cos(-90^\circ) = \boxed{0}$

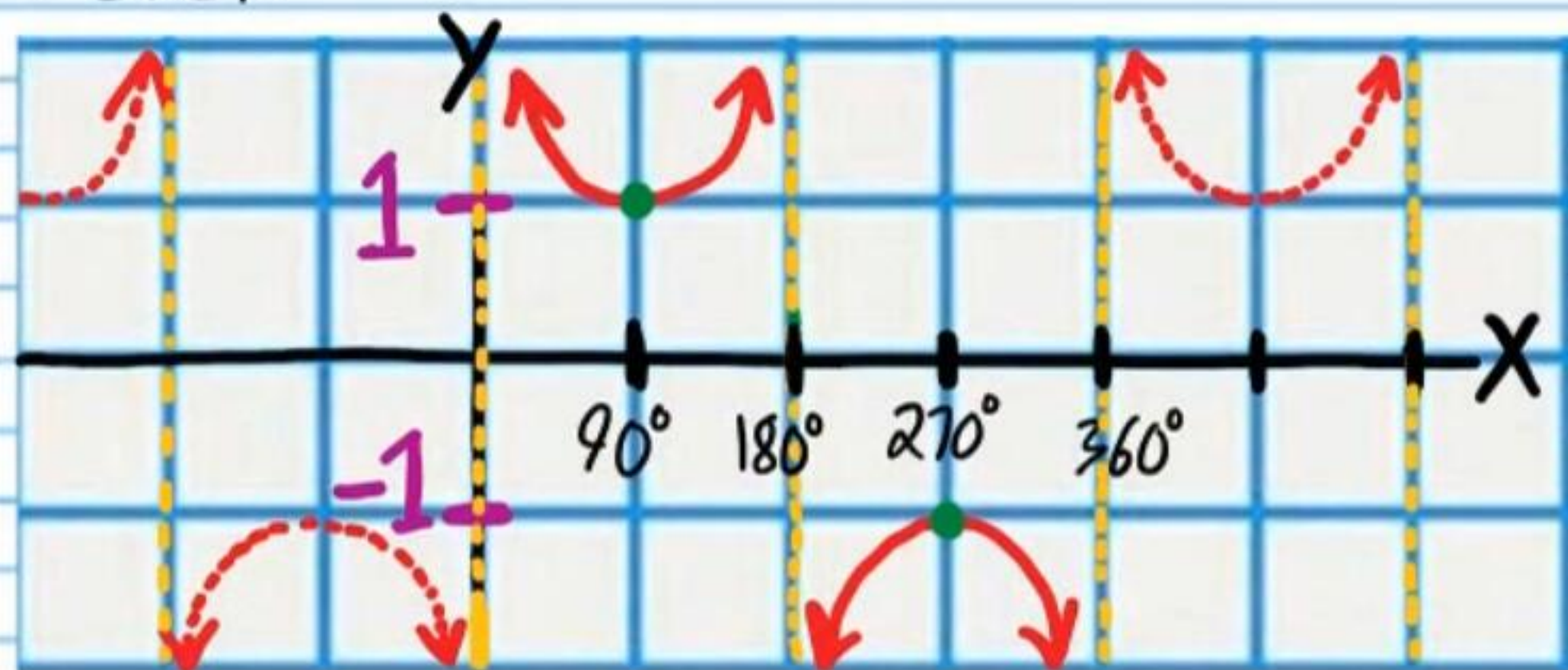
$$y = \sec x$$



Use the graph above to find the following values.

- (21) $\sec(180^\circ) = \boxed{-1}$ (22) $\sec 90^\circ = \boxed{\text{Und}}$ (23) $\sec 270^\circ = \boxed{\text{Und}}$
 (24) $\sec(-90^\circ) = \boxed{\text{Und}}$ (25) $\sec 0^\circ = \boxed{1}$ (26) $\sec(-180^\circ) = \boxed{-1}$

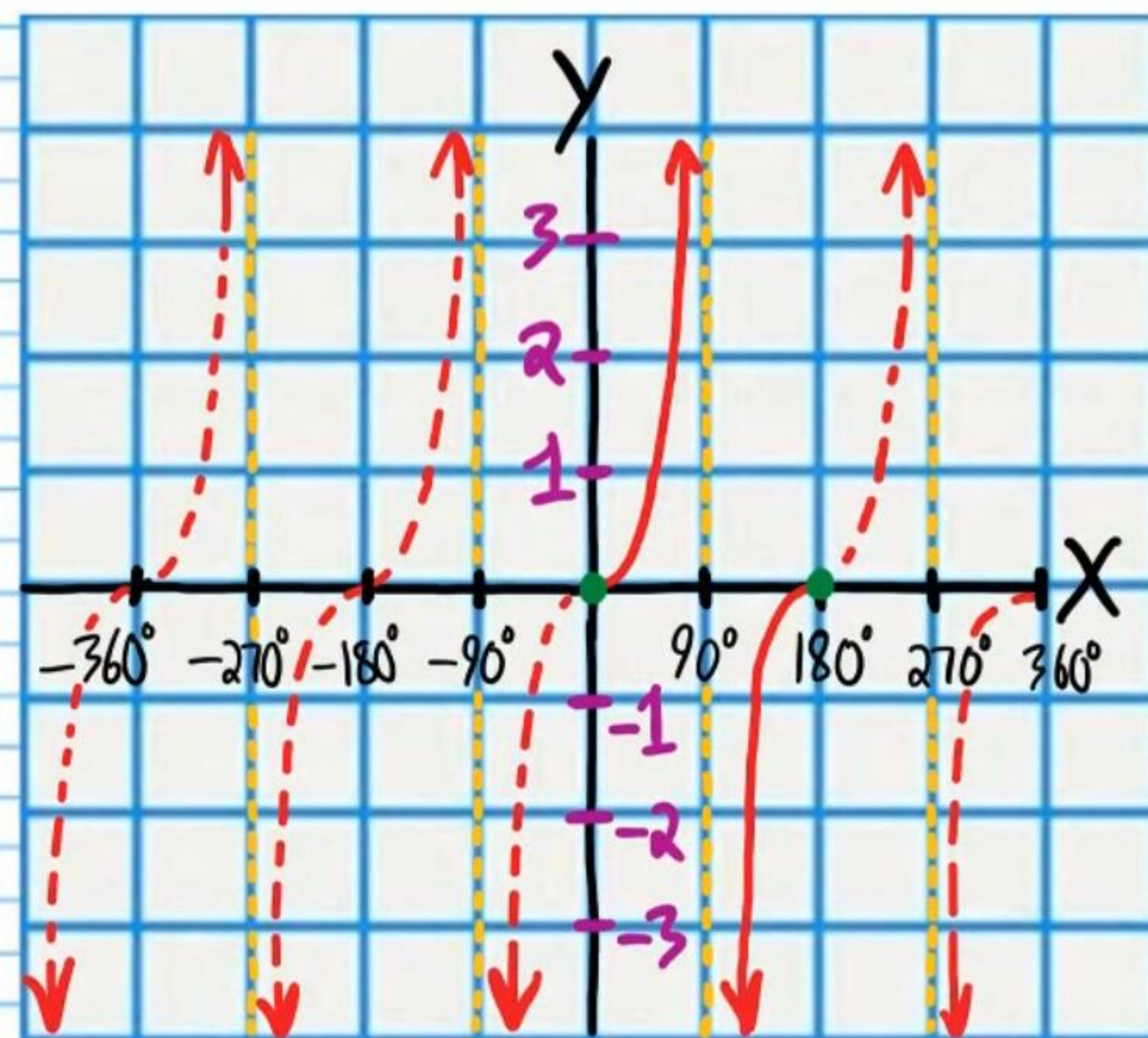
$$y = \csc x$$



Use the graph above to find the following values.

- (27) $\csc 90^\circ = \boxed{1}$ (28) $\csc 180^\circ = \boxed{\text{Und}}$ (29) $\csc(270^\circ) = \boxed{-1}$
 (30) $\csc 0^\circ = \boxed{\text{Und}}$ (31) $\csc(-90^\circ) = \boxed{-1}$ (32) $\csc 360^\circ = \boxed{\text{Und}}$

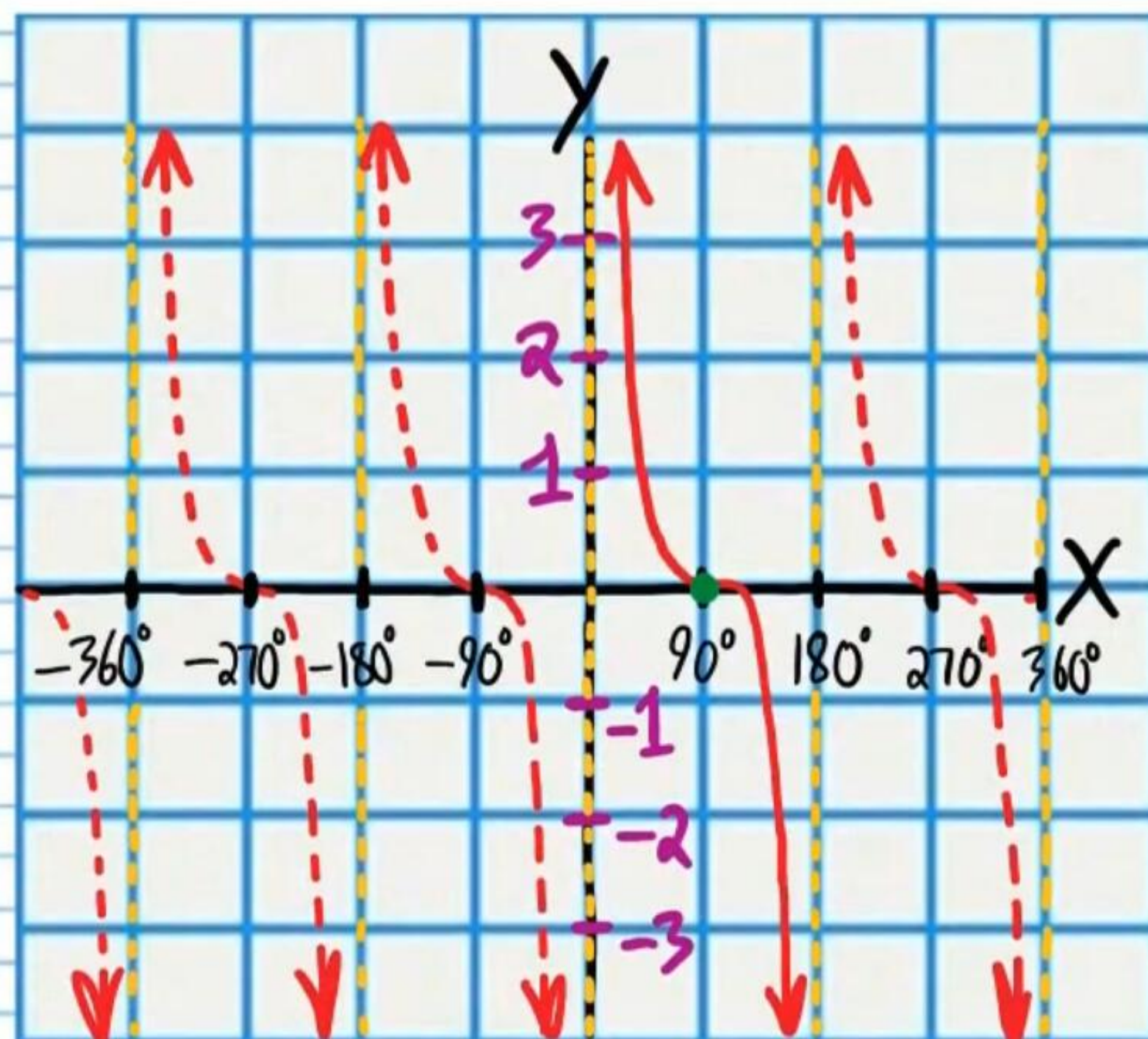
$$y = \tan x$$



Use the graph above to find the following values.

- (33) $\tan 180^\circ = \boxed{0}$ (34) $\tan 90^\circ = \boxed{\text{Und}}$ (35) $\tan 0^\circ = \boxed{0}$
 (36) $\tan(-270^\circ) = \boxed{\text{Und}}$ (37) $\tan(-180^\circ) = \boxed{0}$
 (38) $\tan(360^\circ) = \boxed{0}$

$$y = \cot x$$



Use the graph above to find the following values.

(39) $\cot 90^\circ = \boxed{0}$

(40) $\cot 180^\circ = \boxed{\text{Unh}}$

(41) $\cot(-90^\circ) = \boxed{0}$

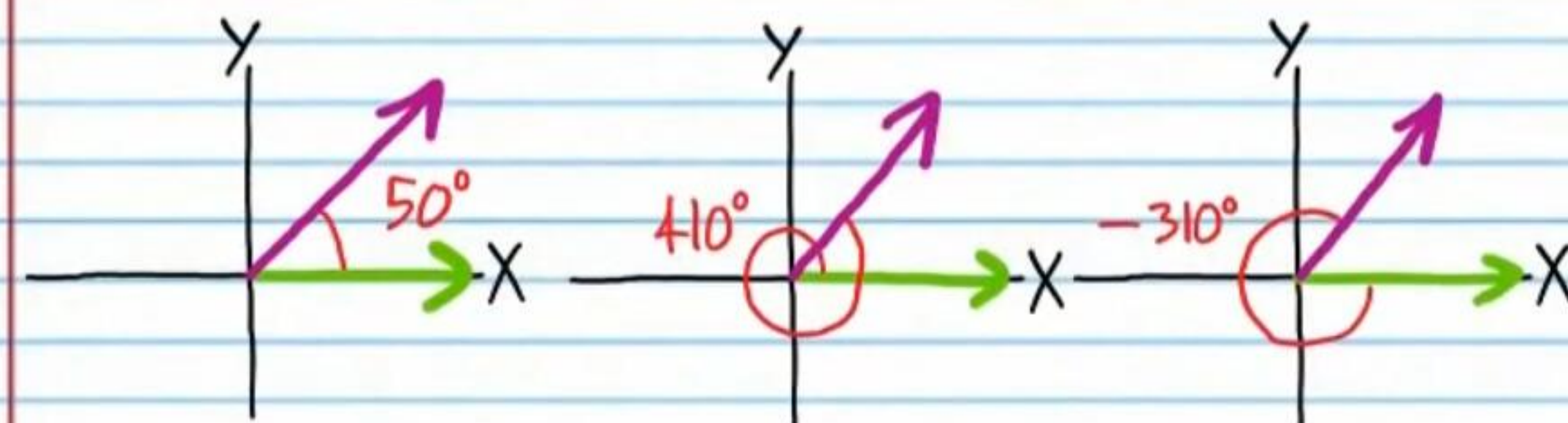
(42) $\cot 0^\circ = \boxed{\text{Und}}$

(43) $\cot 270^\circ = \boxed{0}$

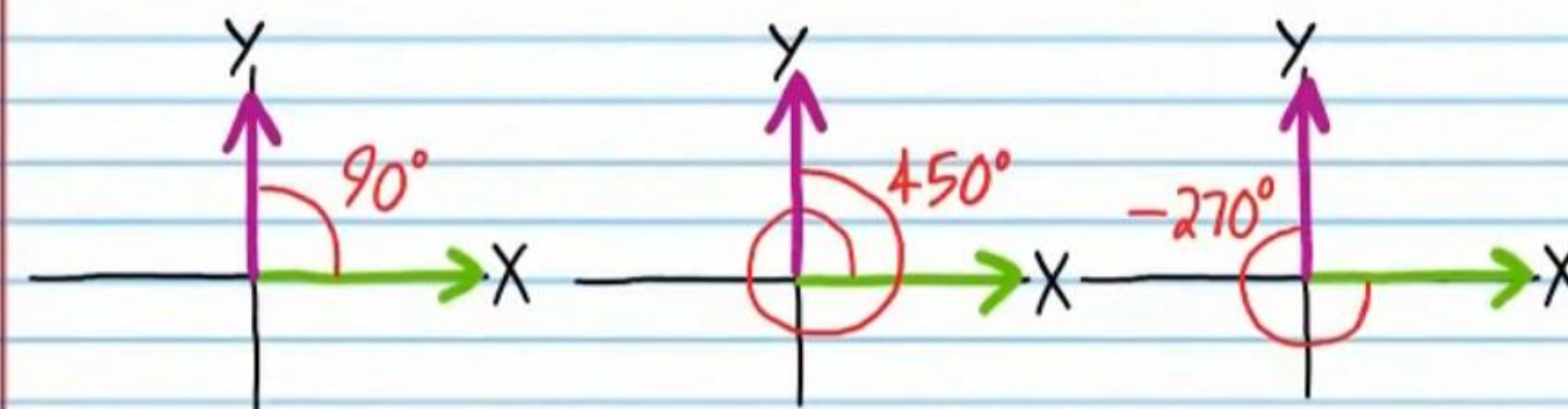
Coterminal Angles

Coterminal Angles: Two or more angles in standard position whose terminal sides have the same location.

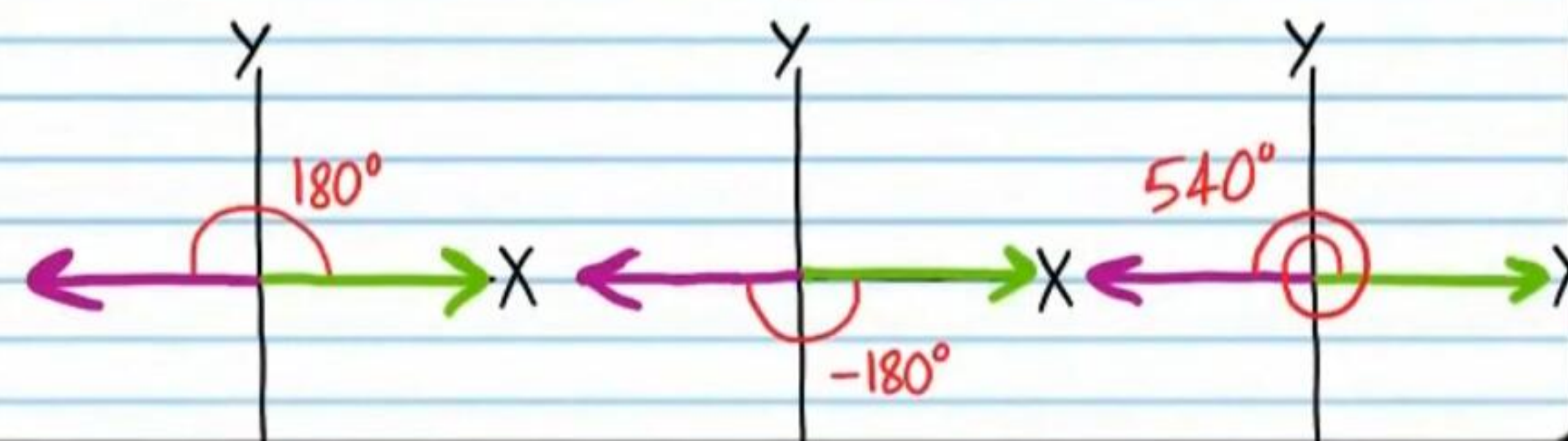
Example I



Example II



Example III



Practice

The measures of angles in standard position are written below. Find the measures of three coterminal angles in standard position.

• (44) 80°

$$\theta_1 = 80 + 360 = \boxed{440^\circ}$$

$$\theta_2 = 80 - 360 = \boxed{-280^\circ}$$

$$\theta_3 = 80 + 360 + 360 = \boxed{800^\circ}$$

(45) 0°

$$\theta_1 = 0 + 360 = \boxed{360^\circ}$$

$$\theta_2 = 0 - 360 = \boxed{-360^\circ}$$

$$\theta_3 = 0 + 360 + 360 = \boxed{720^\circ}$$

(46) -235°

$$\theta_1 = -235 + 360 = \boxed{125^\circ}$$

$$\theta_2 = -235 + 360 + 360 = \boxed{485^\circ}$$

$$\theta_3 = -235 - 360 = \boxed{-595^\circ}$$

• (47) 200°

$$\theta_1 = 200 + 360 = \boxed{560^\circ}$$

$$\theta_2 = 200 - 360 = \boxed{-160^\circ}$$

$$\theta_3 = 200 - 360 - 360 = \boxed{-520^\circ}$$

(48) -315°

$$\theta_1 = -315 + 360 = \boxed{45^\circ}$$

$$\theta_2 = -315 + 360 + 360 = \boxed{405^\circ}$$

$$\theta_3 = -315 - 360 = \boxed{-675^\circ}$$

(49) 1°

$$\theta_1 = 1 + 360 = \boxed{361^\circ}$$

$$\theta_2 = 1 - 360 = \boxed{-359^\circ}$$

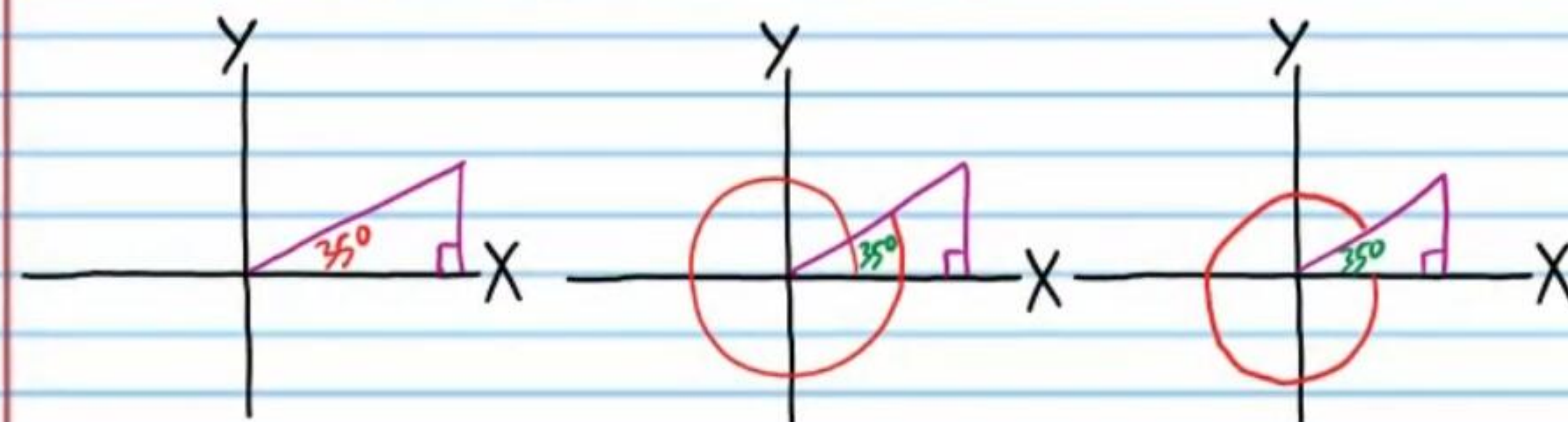
$$\theta_3 = 1 + 360 + 360 = \boxed{721^\circ}$$

Trigonometric Values of Coterminal Angles

Use a calculator to show that the following statements are true. Then, draw a reference triangle in standard position to show why each statement is true.

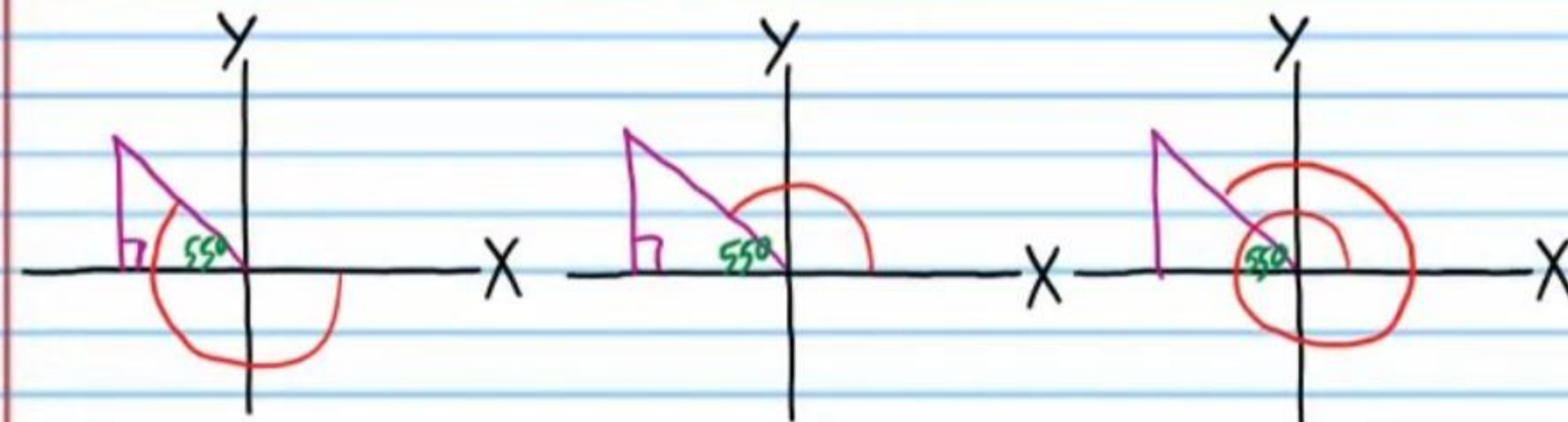
(50) $\sin 35^\circ = \sin 395^\circ = \sin(-325^\circ)$

$$0.5736 \approx 0.5736 \approx 0.5736$$



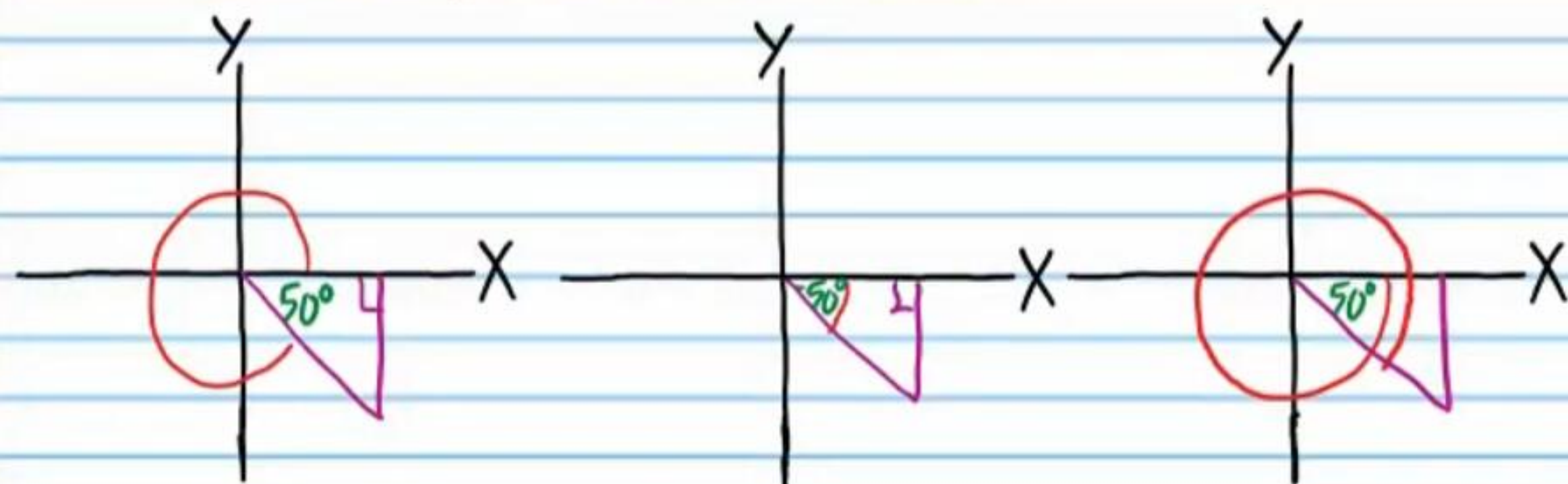
(51) $\tan(-235^\circ) = \tan 125^\circ = \tan 485^\circ$

$$-1.4281 \approx -1.4281 \approx -1.4281$$



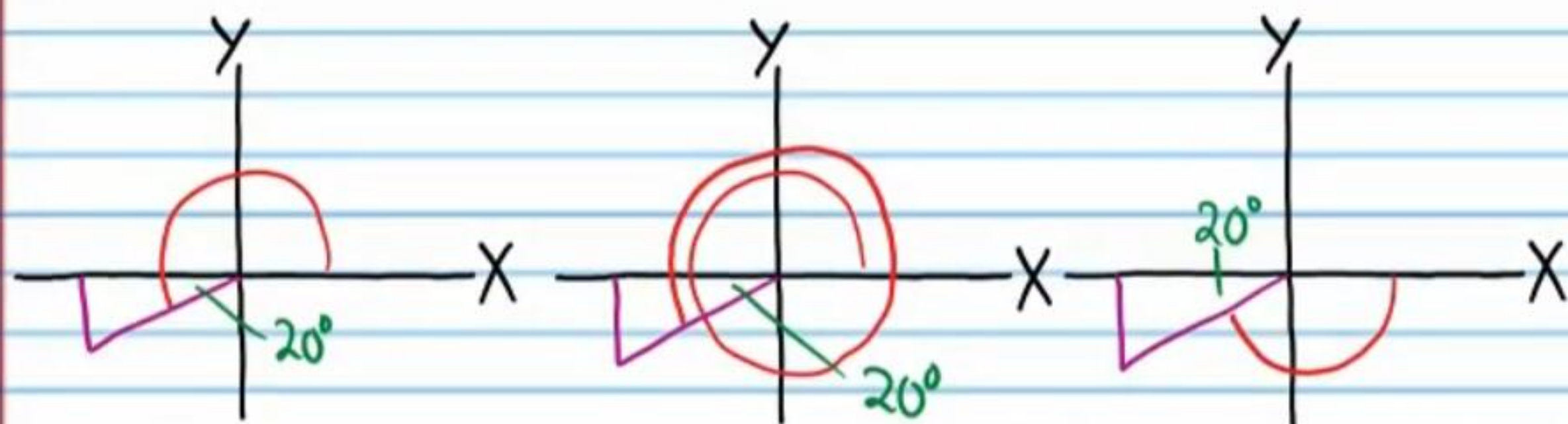
• (52) $\cos 310^\circ = \cos(-50^\circ) = \cos(-410^\circ)$

$0.6428 \approx 0.6428 \approx 0.6428$



• (53) $\sin 200^\circ = \sin 560^\circ = \sin(-160^\circ)$

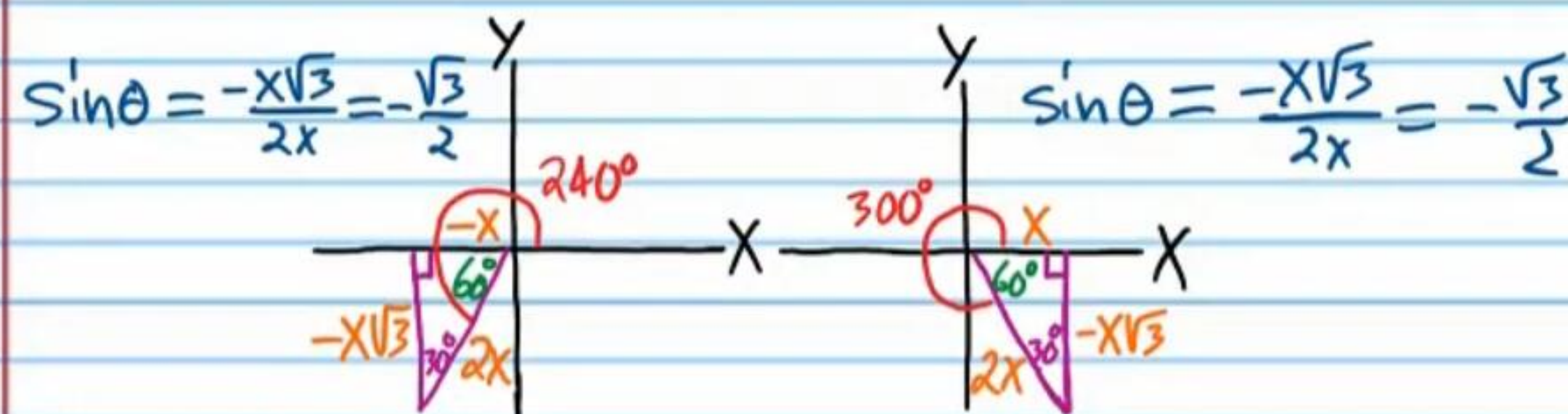
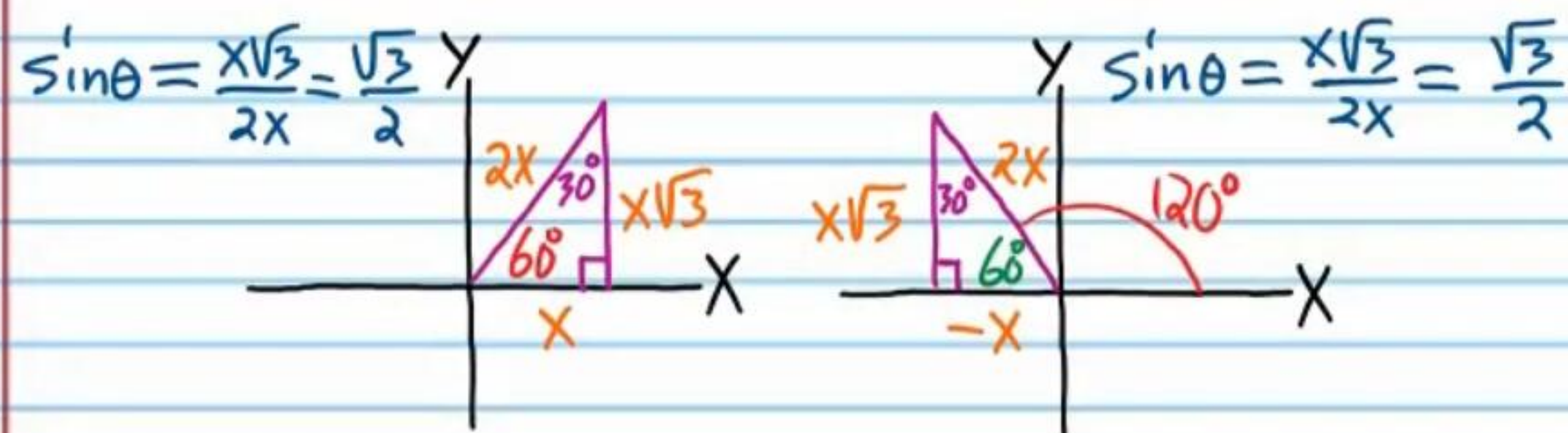
$-0.3420 \approx -0.3420 \approx -0.3420$



Finding Angles That Correspond to Trigonometric Ratios

• (54) If $\sin \theta = -\frac{\sqrt{3}}{2}$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ .

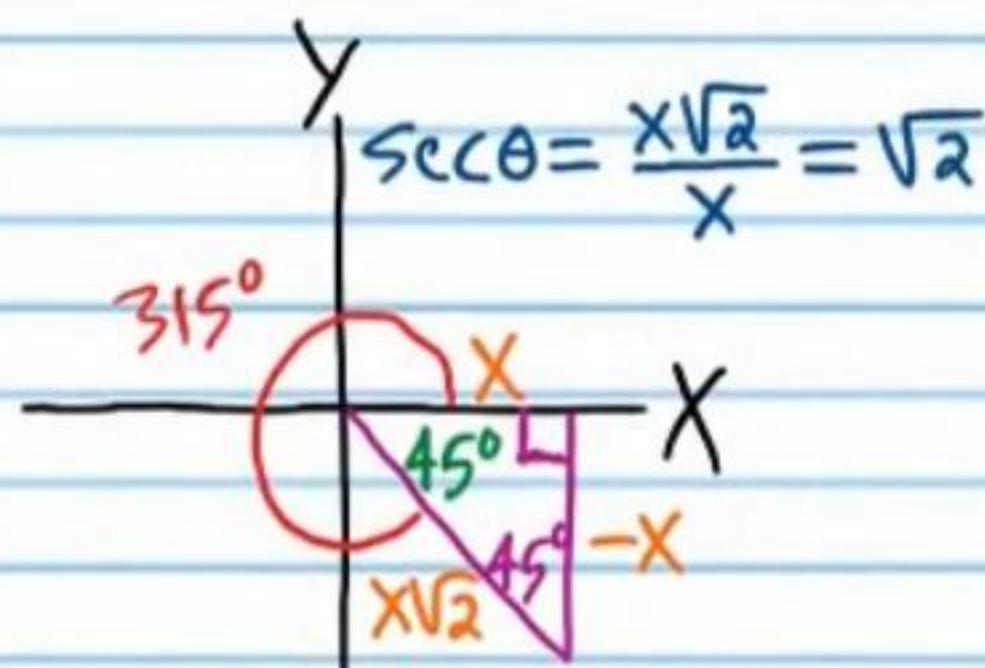
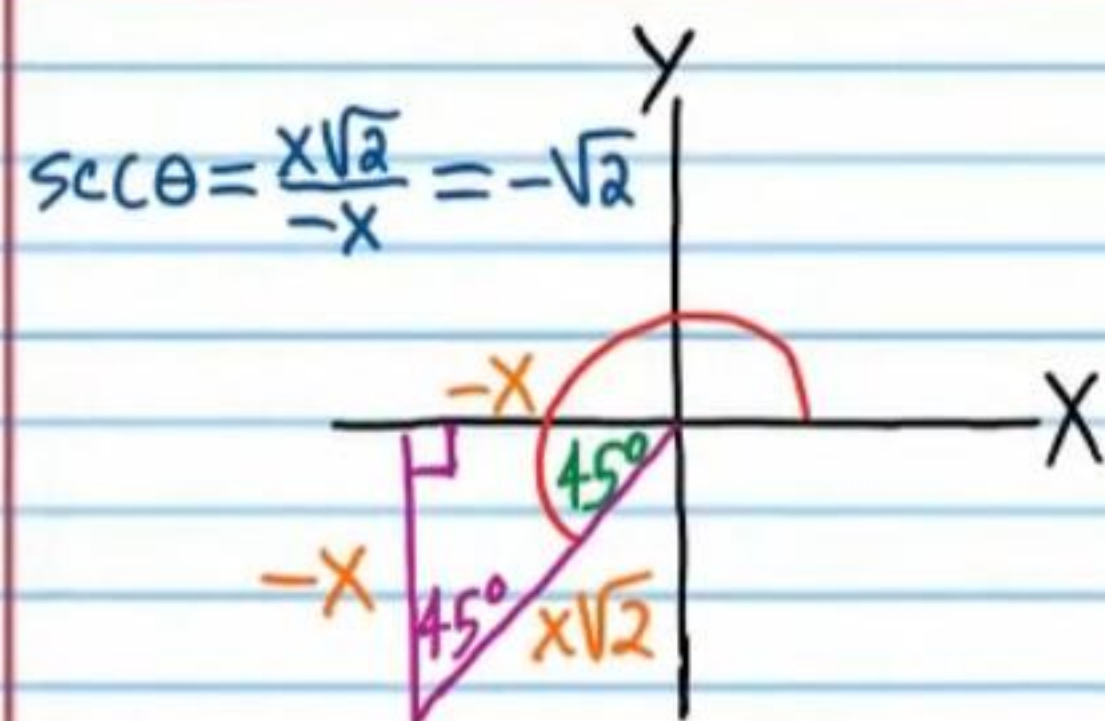
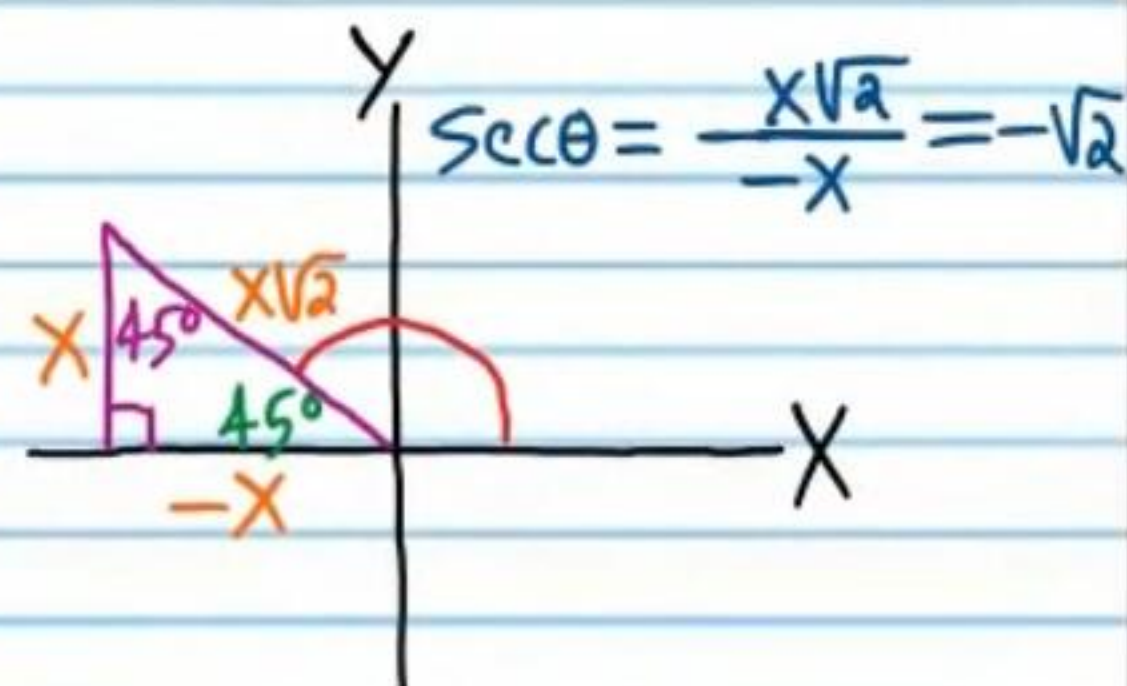
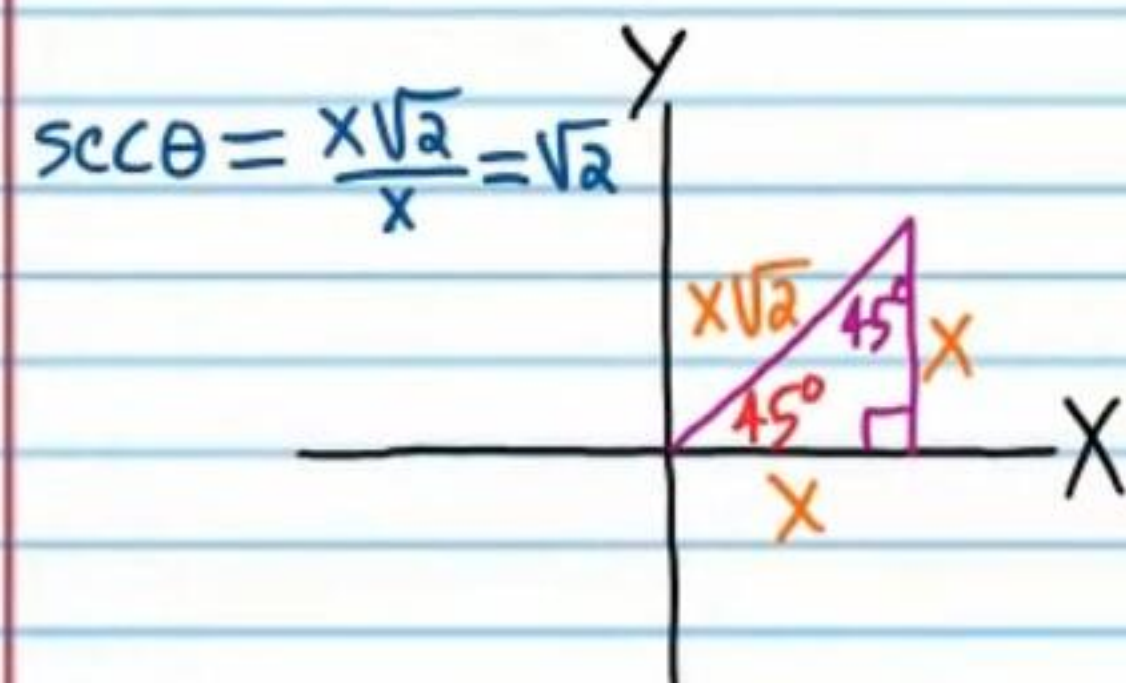
$\sin \theta = -\frac{\sqrt{3}}{2} = -\frac{x\sqrt{3}}{2x}$



$\theta = 240^\circ, 300^\circ$

55) If $\sec\theta = \sqrt{2}$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ .

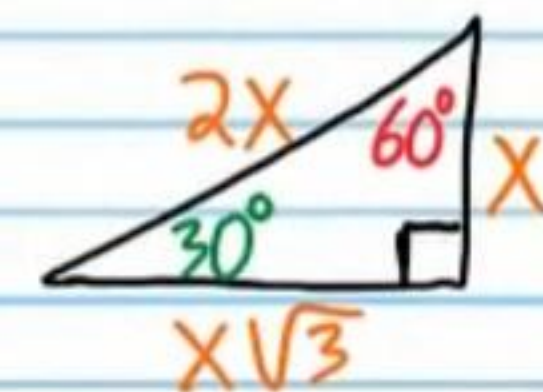
$$\sin\theta = \frac{\sqrt{2}}{1} = \frac{X\sqrt{2}}{X}$$



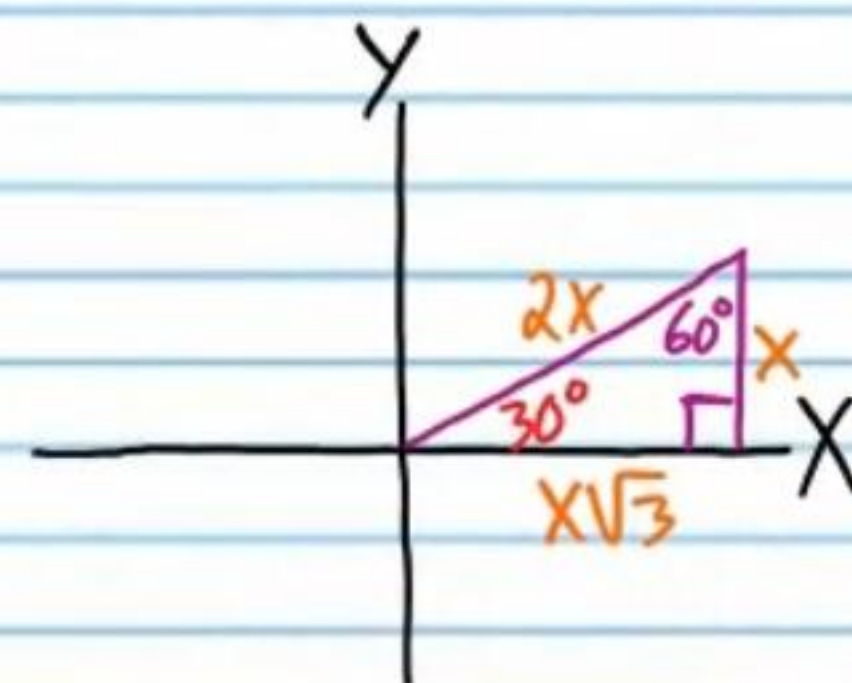
$$\theta = 45^\circ, 315^\circ$$

56) If $\tan\theta = -\frac{1}{\sqrt{3}}$, and $0^\circ < \theta \leq 360^\circ$, Find the values of θ .

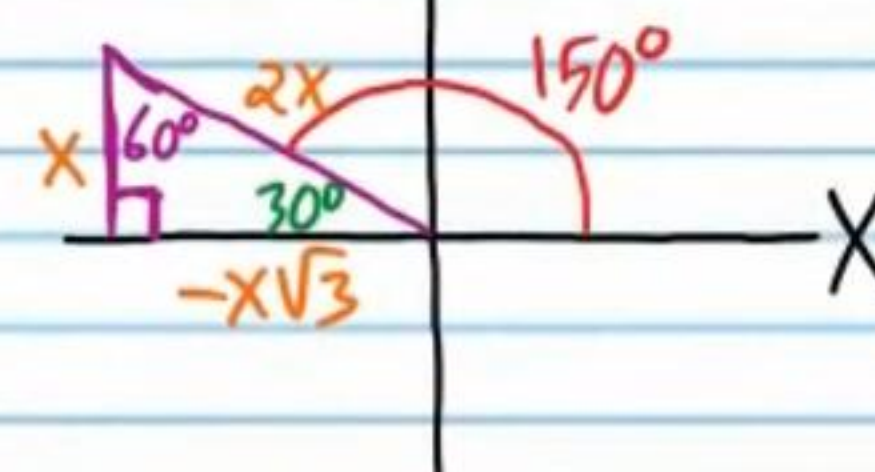
$$\tan\theta = -\frac{1}{\sqrt{3}} = -\frac{X}{X\sqrt{3}}$$



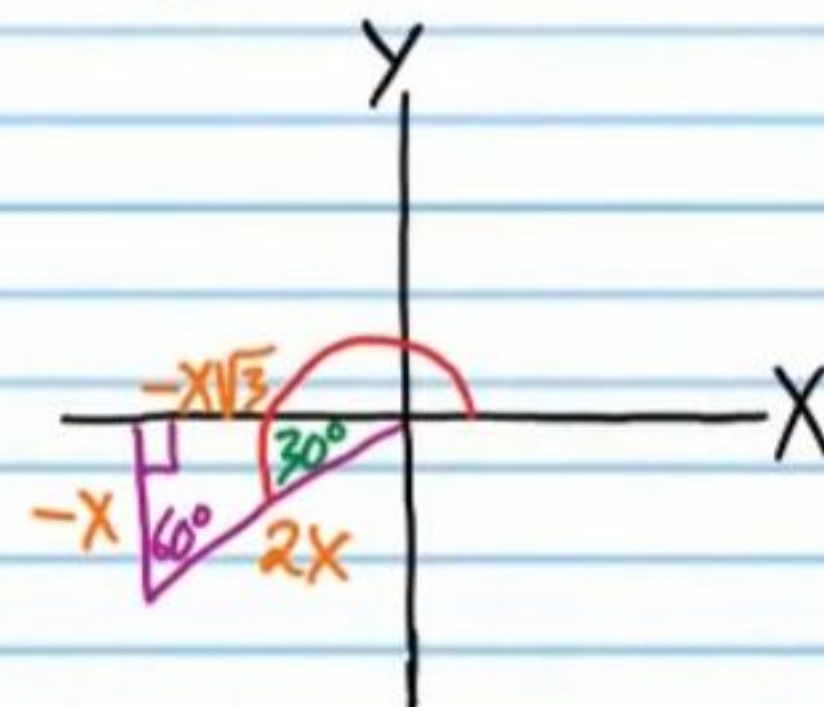
$$\tan\theta = \frac{X}{X\sqrt{3}} = \frac{1}{\sqrt{3}}$$



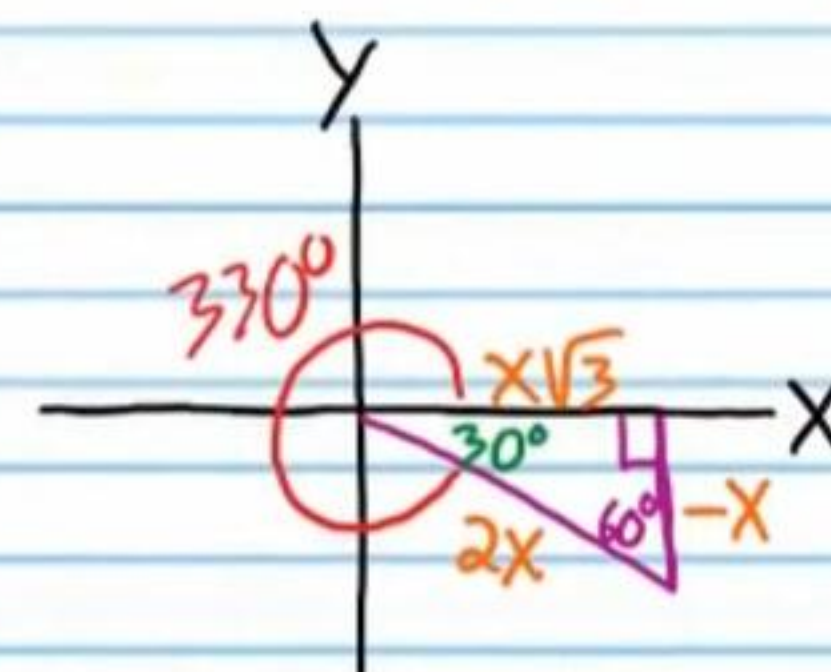
$$\tan\theta = \frac{X}{-X\sqrt{3}} = -\frac{1}{\sqrt{3}}$$



$$\tan\theta = \frac{-X}{-X\sqrt{3}} = \frac{1}{\sqrt{3}}$$



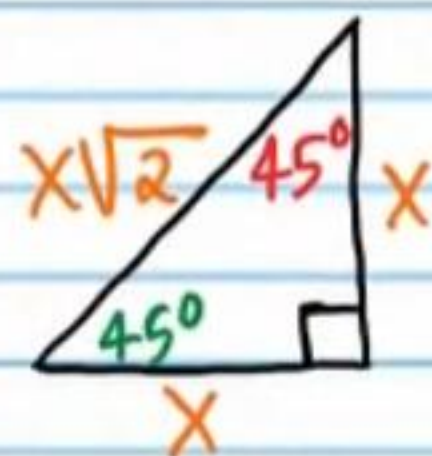
$$\tan\theta = \frac{-X}{X\sqrt{3}} = -\frac{1}{\sqrt{3}}$$



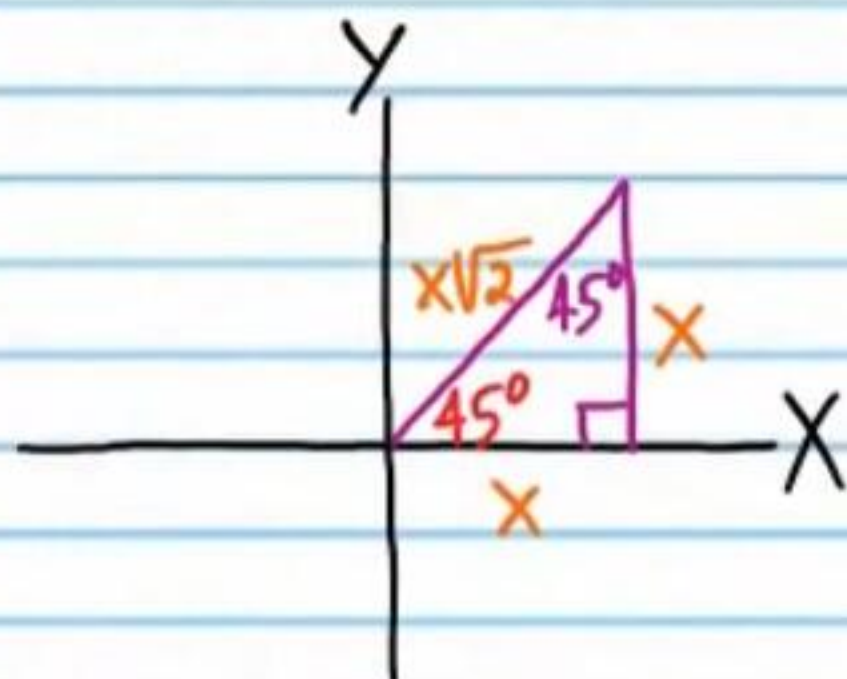
$$\theta = 150^\circ, 330^\circ$$

- (57) If $\cos \theta = \frac{1}{\sqrt{2}}$, and $0^\circ < \theta \leq 360^\circ$, find the values of θ .

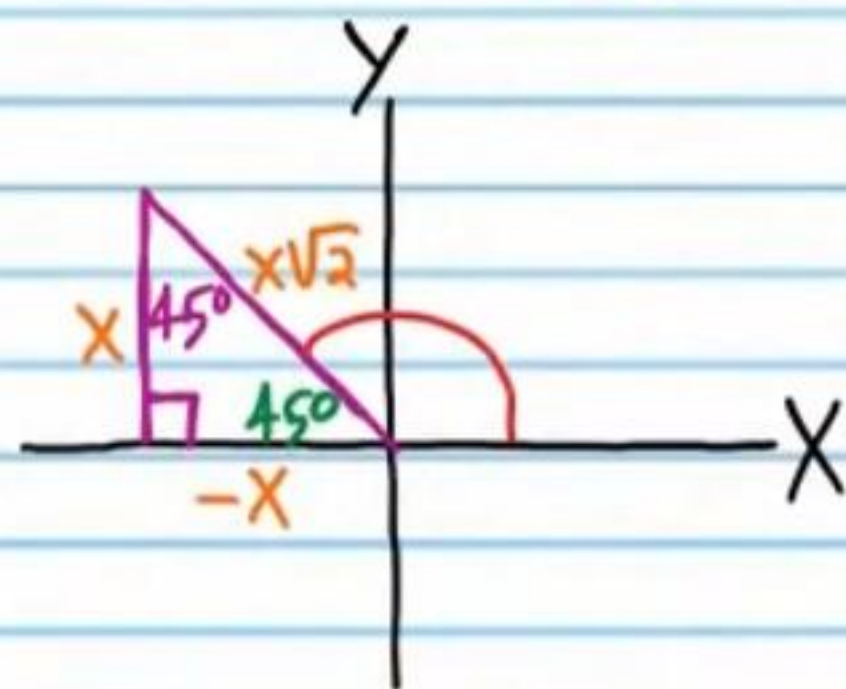
$$\cos \theta = \frac{1}{\sqrt{2}} = \frac{x}{x\sqrt{2}}$$



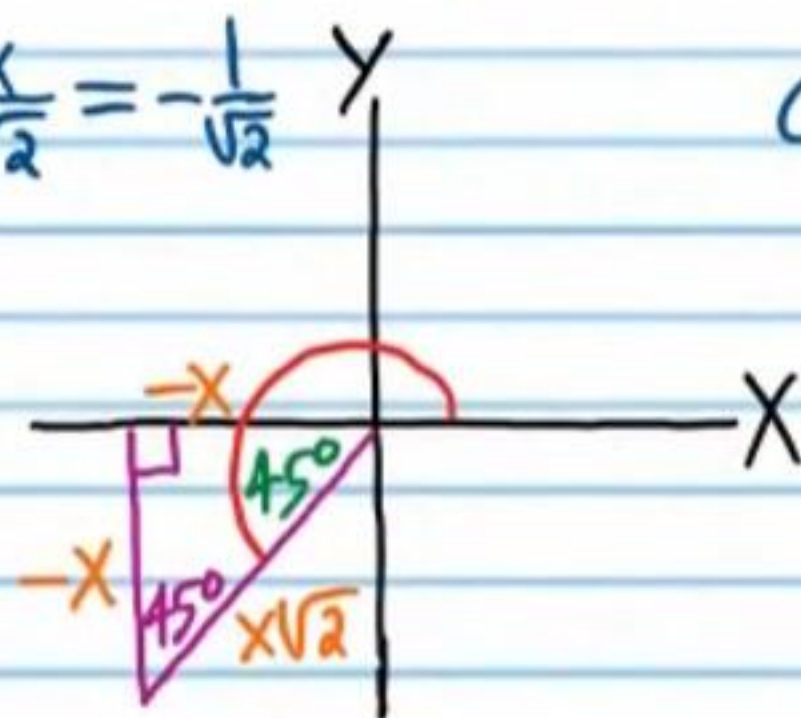
$$\cos \theta = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}}$$



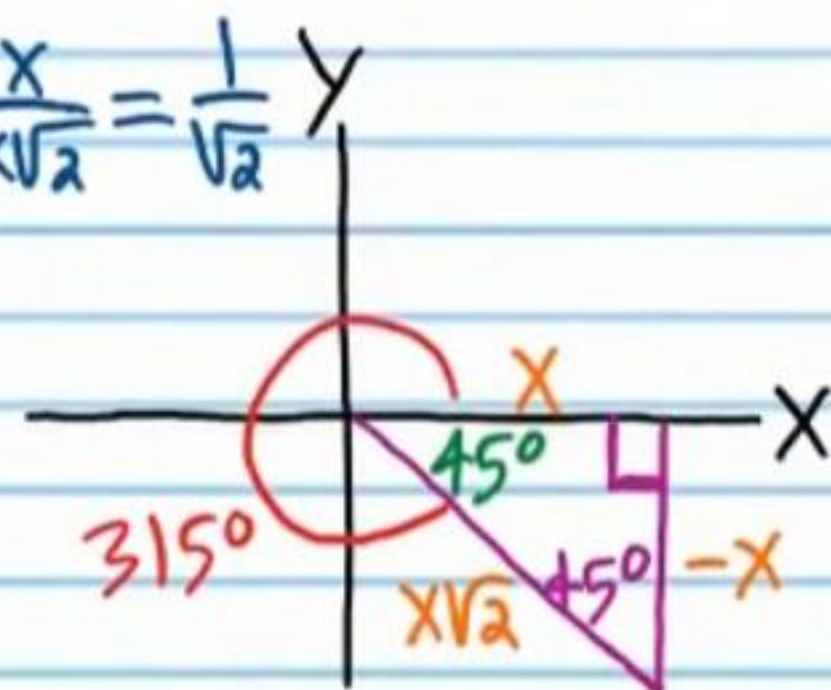
$$\cos \theta = \frac{-x}{x\sqrt{2}} = -\frac{1}{\sqrt{2}}$$



$$\cos \theta = \frac{-x}{x\sqrt{2}} = -\frac{1}{\sqrt{2}}$$



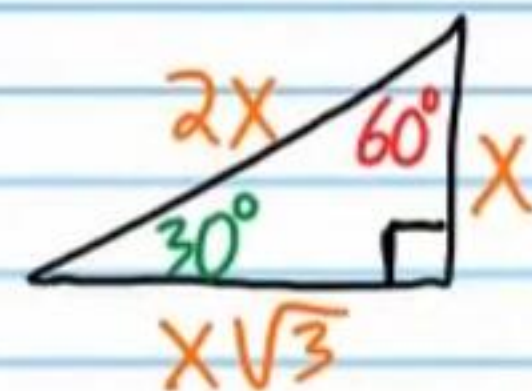
$$\cos \theta = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}}$$



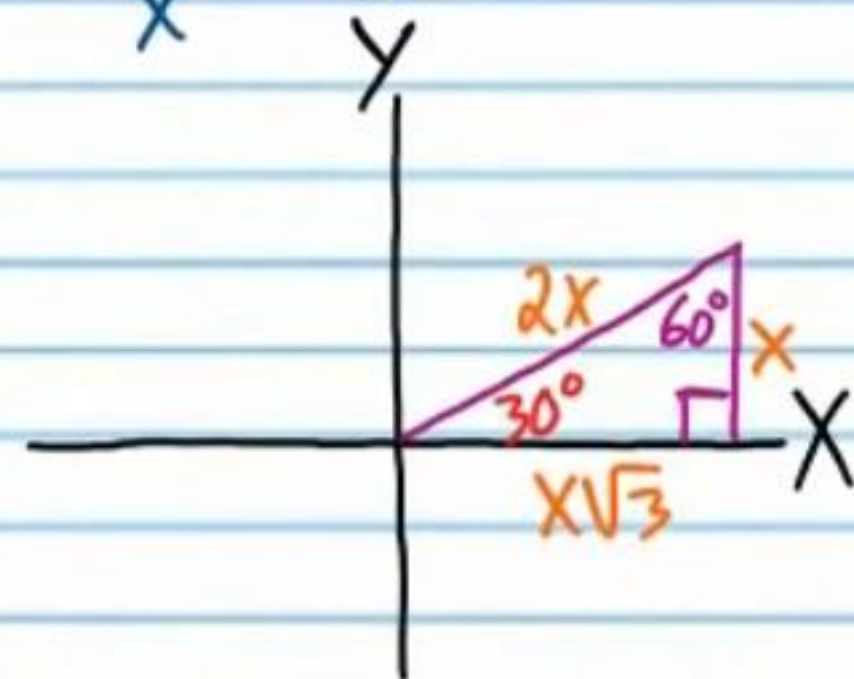
$$\theta = 45^\circ, 315^\circ$$

- (58) If $\csc \theta = -2$, and $0^\circ < \theta \leq 360^\circ$, find the values of θ .

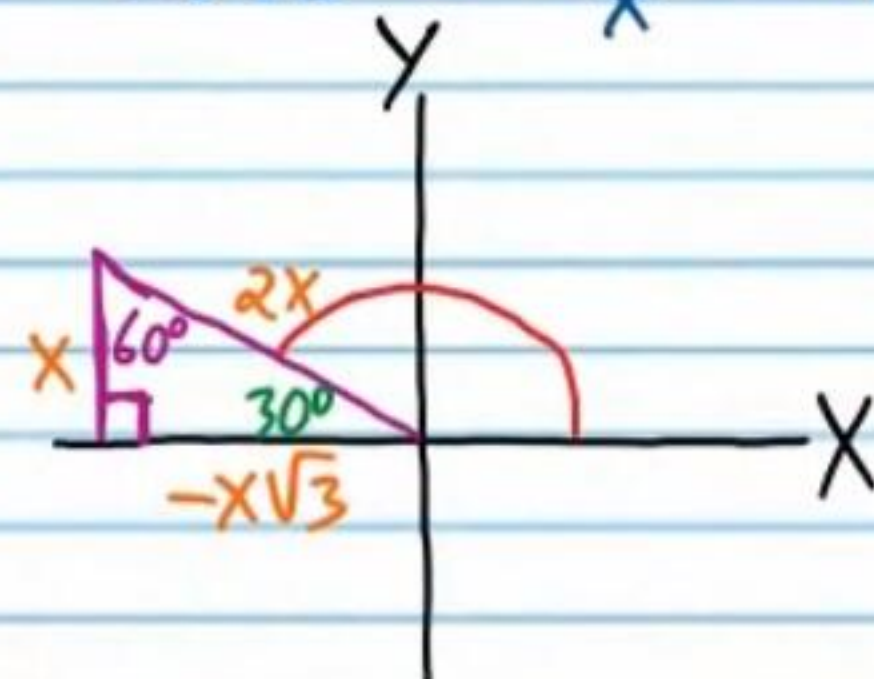
$$\csc \theta = -\frac{2}{1} = \frac{2x}{-x}$$



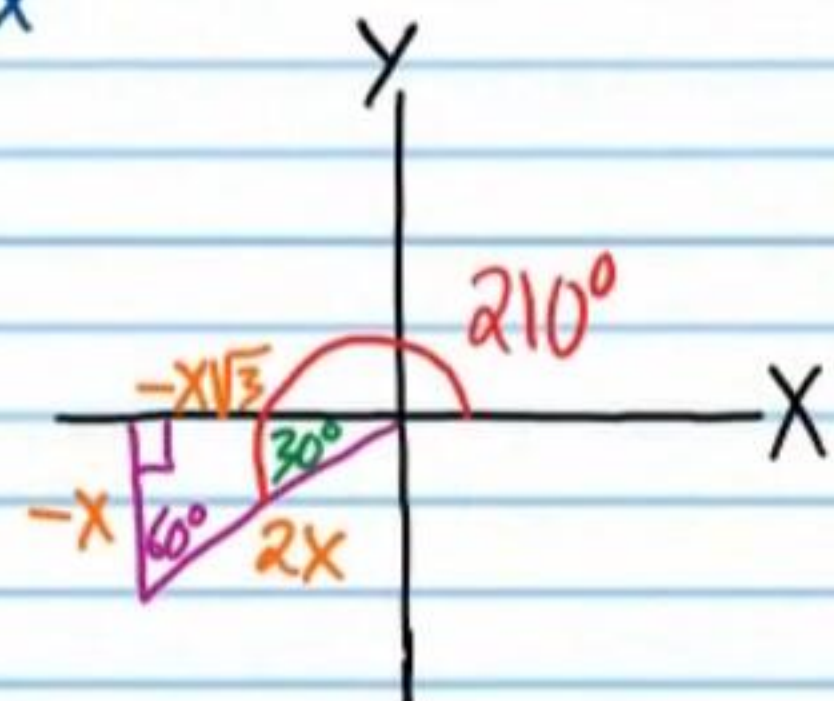
$$\csc \theta = \frac{2x}{x} = 2$$



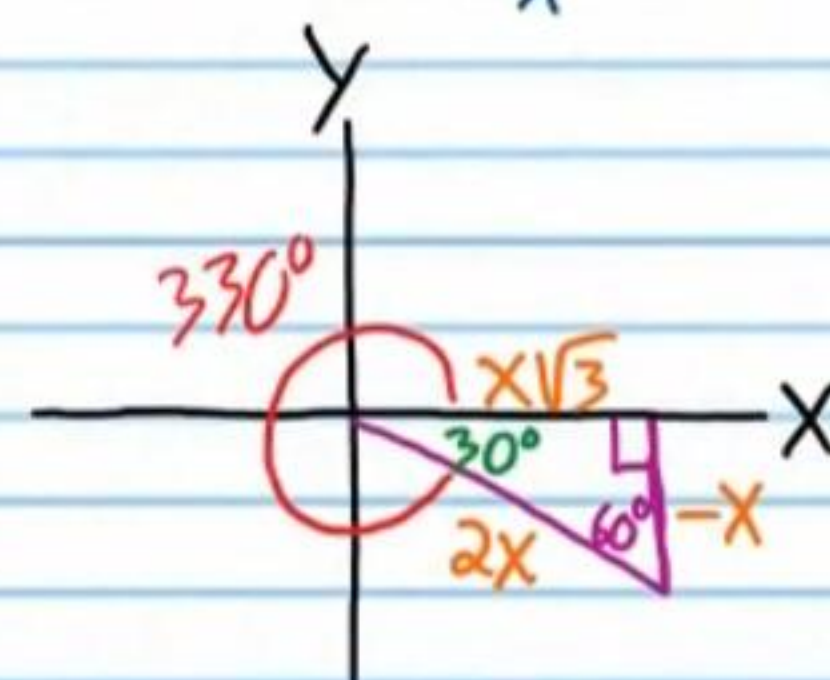
$$\csc \theta = \frac{2x}{x} = 2$$



$$\csc \theta = \frac{2x}{-x} = -2$$



$$\csc \theta = \frac{2x}{-x} = -2$$



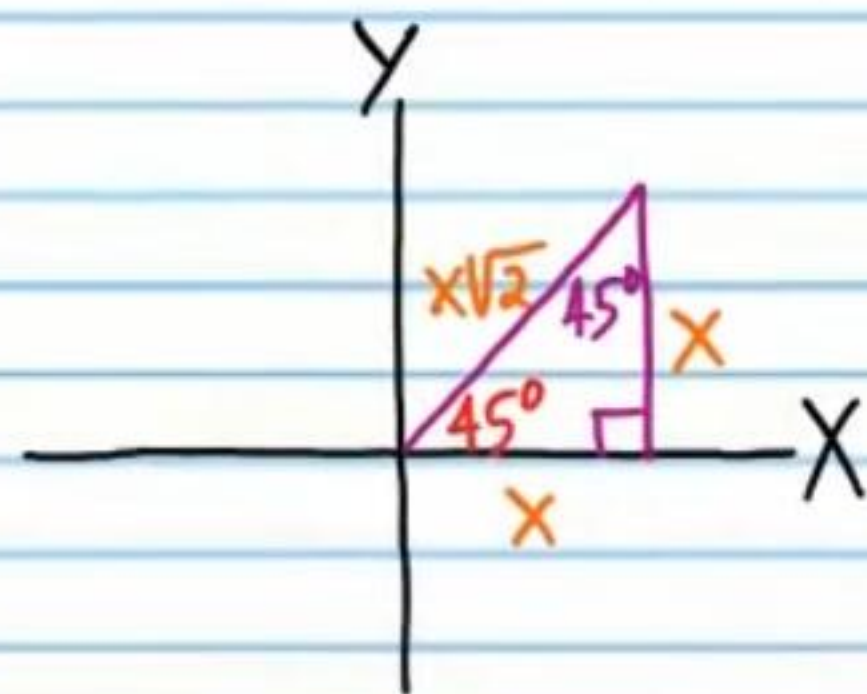
$$\theta = 210^\circ, 330^\circ$$

⑤9 If $\cot \theta = 1$, and $0^\circ < \theta \leq 360^\circ$, find the values of θ .

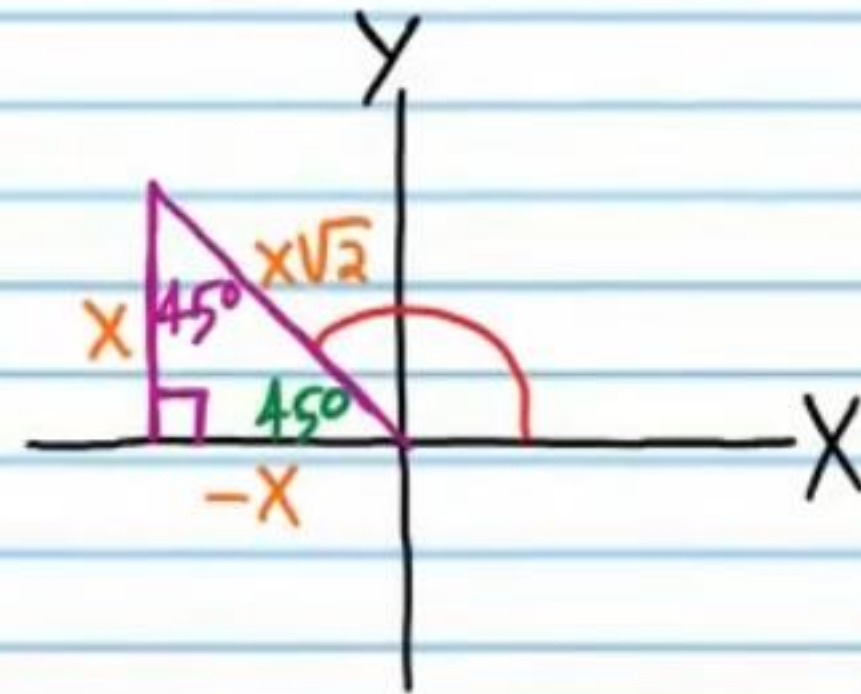
$$\cot \theta = \frac{1}{1} = \frac{x}{x}$$



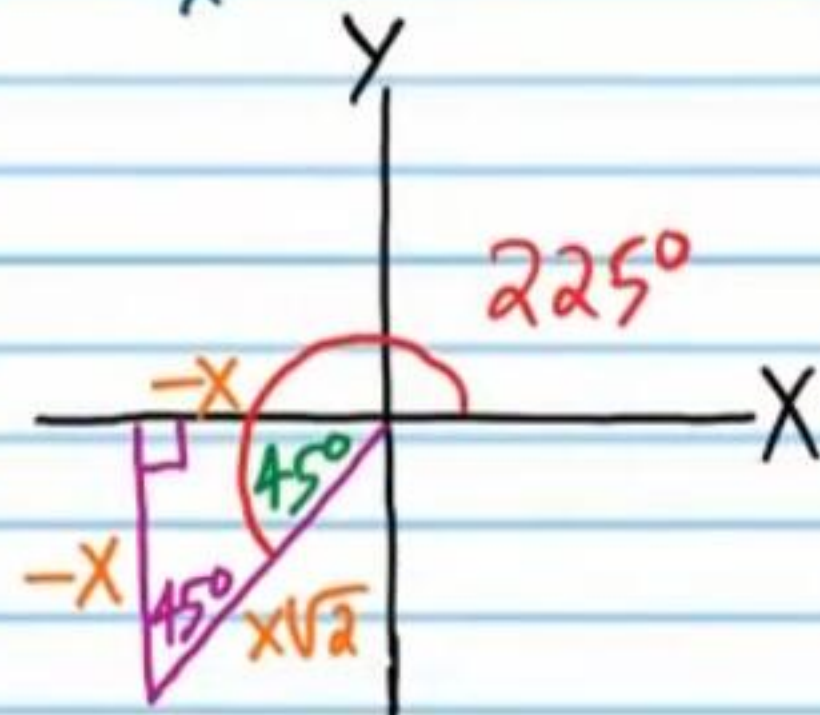
$$\cot \theta = \frac{x}{x} = 1$$



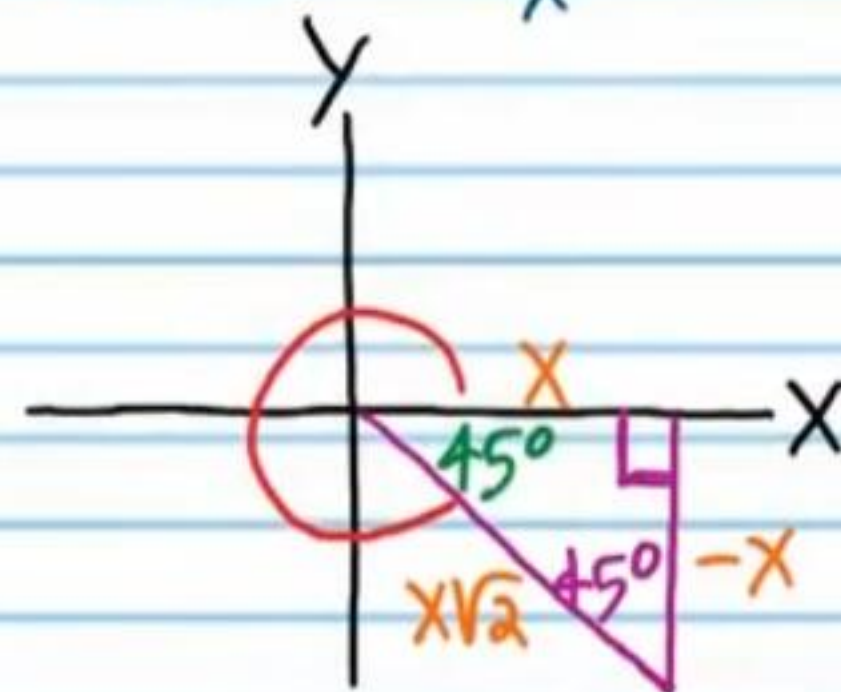
$$\cot \theta = \frac{-x}{x} = -1$$



$$\cot \theta = \frac{-x}{-x} = 1$$



$$\cot \theta = \frac{x}{-x} = -1$$



$$\theta = 45^\circ, 225^\circ$$